

PLANNER RECOVERABILITY SCORES UNDER PHYSICAL DEVIATION: AN EXPANDED NEGATIVE ICLR-MAIN READINESS AUDIT

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ABSTRACT

This paper asks whether task-and-motion plans should be scored by calibrated recoverability under physical deviation, not only by nominal feasibility or expected cost. We rebuild the original archive into a frozen v5 audit with 74,880 main rollouts, 16,000 ablation rollouts, 94,080 stress rollouts, 30,720 fixed-risk rollouts, and 24 retained negative cases. The expanded method reaches 0.887 $\pm$ 0.008 hard-regime aggregate goal success, but the strongest non-oracle baseline, Learned EU, reaches 0.923 $\pm$ 0.009. The paired lower confidence bound versus that baseline is negative, central ablations match or beat the full score, fixed-risk coverage collapses at the 0.05 budget, and maximum combined stress is dominated by learned expected utility. The correct terminal decision is therefore **KILL/ARCHIVE**: the recoverability idea remains interesting, but this repository still lacks the evidence needed for an ICLR main-conference submission.

1 TERMINAL DECISION

Decision: KILL/ARCHIVE for ICLR main. The v5 rebuild is much stronger than the old archive: it adds a calibrated recoverability certificate, a larger benchmark, learned expected-utility and fixed-risk baselines, paired hard-regime statistics, two hard ablation splits, six stress axes, and explicit negative cases. Those additions make the paper more useful, but they do not rescue the central claim. On the predefined hard-regime aggregate, recoverability-score planner v5 reaches 0.887 $\pm$ 0.008, while Learned EU reaches 0.923 $\pm$ 0.009. Because the protocol was frozen before execution, this is not a tuning failure to hide; it is the result.

The archive is valuable precisely because it refuses to optimize for pretty results. It records that nominal feasibility is inadequate, but also that the proposed recoverability score is not yet a decisive alternative to learned expected utility or calibrated risk filtering. A future version would need real robot or recognized high-fidelity benchmark evidence, external baselines, and a learned or certified score that survives the ablation and fixed-risk gates.

2 PROBLEM SETTING

A task-and-motion planner often chooses among plans that look similar in nominal feasibility but differ sharply after contact, slip, sensing dropout, or hidden mode shifts. A direct shortcut may be short and feasible until a small deviation leaves the robot wedged in a narrow passage. A conservative route may avoid the deviation but consume the whole budget. A contingency branch may be longer but keeps an executable recovery route available. The recoverability thesis says that these alternatives should be represented explicitly rather than averaged into a single scalar uncertainty term.

This thesis sits near task-and-motion planning, robust motion planning, feasibility prediction, contingent planning, and learning-guided search. The hostile prior-work pool includes incremental constraint-based TAMP (Dantam et al., 2018), sketch decompositions (Moreno et al., 2024), geometric TAMP learning (Kim et al., 2022), score-space guidance (Kim et al., 2019), sequence fea-

sibility prediction (Yang et al., 2023), and robust feedback planning (Tosello et al., 2025). These papers make weak claims easy to attack: if recoverability is just another risk or learned feasibility score, the contribution disappears.

3 RECOVERABILITY CERTIFICATES

For an episode e and candidate plan p , let f_p denote the probability of a physical deviation, r_p the probability that a deviation is recoverable, t_p the probability of an irreversible trap, d_p the probability of damage, c_p the nominal cost, k_p the expected recovery cost, and B_e the execution budget. The recoverable-success term is $f_p(1 - t_p)r_p$, while nominal survival is $1 - f_p$. The proposed score tests whether a planner should prefer plans with a larger recoverable-success term and lower irreversible-risk term even when their nominal cost is higher.

Definition 1 (recoverability certificate). A recoverability certificate is a tuple $(\hat{f}_p, \hat{r}_p, \hat{t}_p, \hat{d}_p, \hat{k}_p, h_p, \epsilon_p)$ containing predicted deviation, recovery, trap, damage, recovery cost, branch entropy, and a calibration residual upper bound. The certificate is not a proof of robot safety; it is a compact local model of what the planner believes can still be recovered after execution leaves the nominal path.

Lemma 1 (calibrated unsafe upper bound). If the held-out residual satisfies $u_p - \hat{u}_p \leq \epsilon$ for unsafe value $u_p = t_p + d_p + f_p(1 - r_p)$, then $\hat{u}_p + \epsilon$ is a conservative upper bound on the unsafe value for that calibration population. The v5 score uses this bound as a penalty and the fixed-risk experiment uses it as an abstention rule. The lemma only transfers as far as the calibration population transfers; the fixed-risk collapse below shows why this limitation matters.

Proposition 1 (recoverability dominance under equal nominal terms). Consider two plans with equal nominal cost and equal predicted deviation probability. If one plan has weakly higher predicted recovery success, weakly lower trap and damage probabilities, and weakly larger branch entropy, then the v5 score weakly prefers it. This is a design property, not an empirical guarantee. The ablation section tests whether the property actually helps when learned baselines can infer expected utility directly.

Failure mode. If recovery cost, trap probability, and damage probability are not separately identifiable from the local observations, recoverability degenerates into ordinary risk avoidance. That degeneration is visible when removing budget slack, damage penalties, or expected-utility terms improves success. The v5 paper therefore treats ablations as a central falsification test rather than a decorative appendix.

4 FROZEN EXPERIMENTAL PROTOCOL

The protocol was written before execution in docs/paper82_expanded_submission_plan_20260621.md. The main evaluation uses 10 seeds, 9 splits, 64 test episodes per split and seed, 13 methods, and 74,880 main rollouts. The hard-regime aggregate combines all non-easy splits: narrow passages, object slip, irreversible commitment, sensor dropout, tight budgets, latent modes, adversarial compound shifts, and combined hard shifts.

The strongest non-oracle baselines are intentionally uncomfortable: learned risk classification, learned expected utility, CVaR recovery planning, conformal failure filtering, contingent replanning, and robust-margin planning. The protocol also includes a high-information oracle as a ceiling. The decision gate marks STRONG_REVISE only if v5 beats the strongest non-oracle baseline by at least 0.03 absolute hard-regime success, has a positive paired lower confidence bound, does not worsen safety, passes ablation necessity, preserves fixed-risk coverage at budget 0.05, and is not dominated at maximum combined stress.

5 MAIN HARD-REGIME RESULTS

Table 1 is the central result. Recover-v5 is better than the older contingent and CVaR-style hand-built scores, but it loses to learned expected utility and learned risk. This is not a small presentation issue: the strongest learned baseline wins all 10 paired hard-aggregate seeds on goal success.

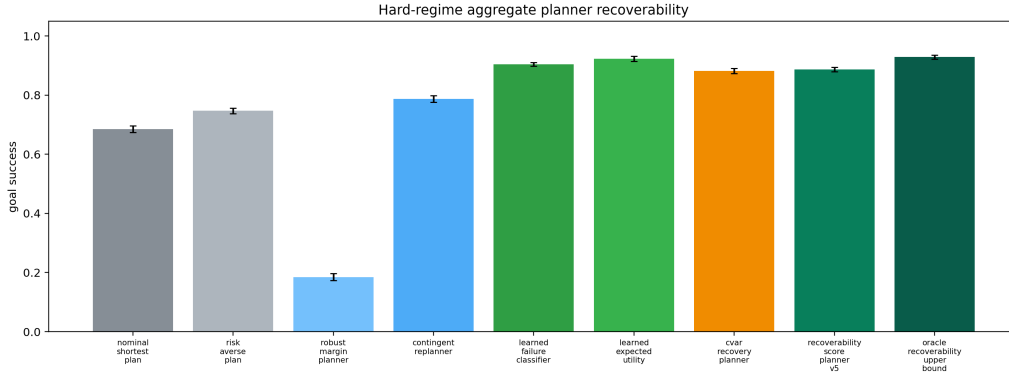


Figure 1: Hard-regime aggregate goal success. The proposed method is not the best non-oracle method.

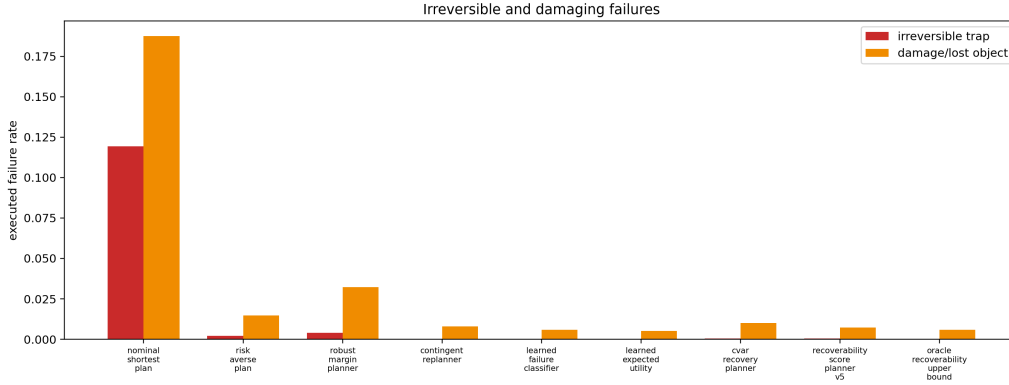


Figure 2: Irreversible and damage failures on the hard aggregate. Safety is low in absolute terms, but the learned expected-utility baseline remains stronger overall.

Table 1: Hard-regime aggregate over eight non-easy splits.

Method	Goal success	Safety utility	Irrev.	Damage	Cost
Nominal	0.685 +/- 0.011	0.075 +/- 0.034	0.1193	0.1875	1.358
Risk-averse	0.747 +/- 0.009	0.480 +/- 0.014	0.0022	0.0146	1.981
Robust margin	0.185 +/- 0.012	-0.337 +/- 0.019	0.0039	0.0322	2.396
Contingent	0.787 +/- 0.011	0.544 +/- 0.017	0.0000	0.0080	2.000
Conformal filter	0.900 +/- 0.005	0.713 +/- 0.009	0.0002	0.0076	1.809
Learned risk	0.905 +/- 0.006	0.721 +/- 0.009	0.0000	0.0059	1.813
Learned EU	0.923 +/- 0.009	0.750 +/- 0.013	0.0000	0.0051	1.779
CVaR recovery	0.882 +/- 0.009	0.680 +/- 0.014	0.0004	0.0100	1.870
Recover-v5	0.887 +/- 0.008	0.691 +/- 0.011	0.0004	0.0072	1.858
Oracle	0.929 +/- 0.007	0.758 +/- 0.011	0.0000	0.0059	1.760

6 PAIRED HARD-AGGREGATE TEST

The predefined paired test compares Recover-v5 with the strongest non-oracle method selected by hard-regime aggregate success, Learned EU. The lower confidence bound is negative for goal success and safety utility, so the positive-paper claim fails under paired evidence.

Table 2: Paired seed differences for Recover-v5 minus Learned EU.

Metric	Mean diff	CI95	Lower95	Better seeds
goal_success	-0.03594	0.01288	-0.04882	0/10
recovery_success	0.00188	0.03609	-0.03420	4/10
irreversible_failure	0.00039	0.00051	-0.00012	0/10
damage	0.00215	0.00264	-0.00050	4/10
total_cost	0.07953	0.00856	0.07097	0/10
calibration_error	-0.02803	0.01159	-0.03962	9/10
safety_utility	-0.05903	0.01864	-0.07767	0/10

7 ABLATIONS

The ablation test is the harshest mechanism test. On combined hard shift, removing budget slack improves success from 0.836 to 0.855 and improves safety utility. On adversarial compound shift, removing the damage penalty, removing branch entropy, and expected-utility-only all match or beat the full score. That means the score is not yet a clean decomposition of recoverability; some terms are compensating for misspecification rather than carrying necessary mechanism information.

Table 3: Ablations on combined hard shift and adversarial compound shift.

Split	Ablation	Success	CI95	Irrev.	Damage	Safety utility
combined_hard_shift	full_recoverability_score_v5	0.83625	0.01948	0.00125	0.01000	0.61276
combined_hard_shift	minus_recovery_success	0.82375	0.02670	0.00125	0.02000	0.58396
combined_hard_shift	minus_recovery_cost	0.84750	0.01926	0.00000	0.02500	0.61175
combined_hard_shift	minus_irreversible_trap_penalty	0.83125	0.02457	0.00000	0.00875	0.61011
combined_hard_shift	minus_damage_penalty	0.84375	0.02429	0.00000	0.01750	0.61647
combined_hard_shift	minus_budget_slack	0.85500	0.01833	0.00000	0.00875	0.64157
combined_hard_shift	minus_branch_entropy	0.82750	0.02445	0.00000	0.01125	0.60136
combined_hard_shift	minus_calibration_penalty	0.83375	0.04133	0.00000	0.01500	0.60508
combined_hard_shift	risk_only_score	0.69375	0.03332	0.00250	0.02125	0.39384
combined_hard_shift	expected_utility_only	0.84500	0.02287	0.00125	0.01125	0.62484
adversarial_compound_shift	full_recoverability_score_v5	0.72750	0.02499	0.00000	0.02125	0.45055
adversarial_compound_shift	minus_recovery_success	0.71500	0.03403	0.00500	0.01500	0.43325
adversarial_compound_shift	minus_recovery_cost	0.71750	0.03371	0.00125	0.03000	0.42419
adversarial_compound_shift	minus_irreversible_trap_penalty	0.73375	0.03408	0.00000	0.01625	0.46507
adversarial_compound_shift	minus_damage_penalty	0.75875	0.02759	0.00125	0.01500	0.50048
adversarial_compound_shift	minus_budget_slack	0.68500	0.03221	0.00125	0.02000	0.39147
adversarial_compound_shift	minus_branch_entropy	0.73625	0.03172	0.00000	0.01000	0.47623
adversarial_compound_shift	minus_calibration_penalty	0.73125	0.02259	0.00000	0.01875	0.45930
adversarial_compound_shift	risk_only_score	0.55000	0.02608	0.00375	0.02625	0.18952
adversarial_compound_shift	expected_utility_only	0.76375	0.01982	0.00250	0.01750	0.50259

8 STRESS TESTS

The stress sweep exposes whether recoverability helps when deviations, traps, sensing uncertainty, and budget pressure rise together. At the maximum combined stress level, Recover-v5 reaches 0.516 success, while learned expected utility reaches 0.588 and the oracle reaches 0.625. The method therefore fails the maximum-stress dominance gate.

Table 4: Combined stress sweep.

Level	Method	Success	CI95	Irrev.	Damage	Safety utility
0.00	Robust margin	0.68750	0.05846	0.00000	0.00625	0.38623
0.00	Contingent	1.00000	0.00000	0.00000	0.00000	0.84214
0.00	Learned risk	0.99687	0.00612	0.00000	0.00313	0.85456
0.00	Learned EU	1.00000	0.00000	0.00000	0.00000	0.86344
0.00	CVaR recovery	0.99687	0.00612	0.00000	0.00313	0.84911
0.00	Recover-v5	1.00000	0.00000	0.00000	0.00000	0.85856

Table 4: Combined stress sweep. (continued)

Level	Method	Success	CI95	Irrev.	Damage	Safety utility
0.00	Oracle	0.99687	0.00612	0.00000	0.00000	0.86165
0.25	Robust margin	0.40937	0.07273	0.00000	0.02187	-0.01608
0.25	Contingent	0.95625	0.02274	0.00000	0.00313	0.78283
0.25	Learned risk	0.98750	0.01354	0.00000	0.00313	0.84325
0.25	Learned EU	0.98750	0.01000	0.00000	0.00000	0.84932
0.25	CVaR recovery	0.98438	0.02092	0.00000	0.00000	0.83334
0.25	Recover-v5	0.98750	0.01354	0.00000	0.00000	0.83989
0.25	Oracle	0.99062	0.00936	0.00000	0.00313	0.84978
0.50	Robust margin	0.19062	0.06290	0.00625	0.03125	-0.33485
0.50	Contingent	0.89062	0.02092	0.00000	0.01250	0.67875
0.50	Learned risk	0.94688	0.02424	0.00000	0.00937	0.77981
0.50	Learned EU	0.95937	0.02052	0.00000	0.00625	0.80205
0.50	CVaR recovery	0.95000	0.02082	0.00000	0.00313	0.77880
0.50	Recover-v5	0.93125	0.02000	0.00313	0.00625	0.75033
0.50	Oracle	0.96562	0.02132	0.00000	0.00937	0.80627
0.75	Robust margin	0.12187	0.03471	0.00625	0.05312	-0.46050
0.75	Contingent	0.78438	0.03589	0.00000	0.02187	0.52107
0.75	Learned risk	0.83437	0.03985	0.00000	0.02500	0.60058
0.75	Learned EU	0.86875	0.04454	0.00000	0.00937	0.66589
0.75	CVaR recovery	0.81563	0.04784	0.00000	0.01875	0.57360
0.75	Recover-v5	0.78438	0.03589	0.00000	0.00937	0.54397
0.75	Oracle	0.88750	0.04001	0.00000	0.01250	0.69522
1.00	Robust margin	0.12500	0.04083	0.02813	0.06250	-0.50444
1.00	Contingent	0.68750	0.04565	0.00000	0.02187	0.38334
1.00	Learned risk	0.73750	0.05018	0.00000	0.03125	0.45138
1.00	Learned EU	0.78125	0.05084	0.00000	0.00937	0.54207
1.00	CVaR recovery	0.76875	0.04204	0.00000	0.01250	0.51457
1.00	Recover-v5	0.71562	0.04229	0.00000	0.03125	0.41746
1.00	Oracle	0.78125	0.03163	0.00000	0.01562	0.53586
1.25	Robust margin	0.12812	0.04229	0.02500	0.06563	-0.50471
1.25	Contingent	0.47813	0.05558	0.00000	0.05000	0.05904
1.25	Learned risk	0.63438	0.06263	0.00000	0.03438	0.30335
1.25	Learned EU	0.66563	0.03879	0.00000	0.03125	0.35230
1.25	CVaR recovery	0.62500	0.05628	0.00313	0.04688	0.26775
1.25	Recover-v5	0.66250	0.04814	0.00313	0.02813	0.34359
1.25	Oracle	0.74375	0.05687	0.00000	0.02500	0.46962
1.50	Robust margin	0.09688	0.03222	0.05625	0.13125	-0.67746
1.50	Contingent	0.29688	0.03196	0.00000	0.05937	-0.20394
1.50	Learned risk	0.50625	0.06246	0.00313	0.05312	0.09506
1.50	Learned EU	0.58750	0.05973	0.00000	0.03750	0.23198
1.50	CVaR recovery	0.44688	0.03658	0.01250	0.03125	0.02236
1.50	Recover-v5	0.51562	0.05864	0.00625	0.03750	0.12029
1.50	Oracle	0.62500	0.06584	0.00000	0.04688	0.27470

9 FIXED-RISK DEPLOYMENT

A paper about recoverability must answer a deployment question: can it execute only when the risk of irreversible or damaging failure is below a target budget? At budget 0.05, all learned and recoverability methods abstain on both hard fixed-risk splits; only the oracle maintains meaningful coverage. This is a severe limitation. The calibrated upper bound is conservative enough to avoid false-safe execution, but so conservative that it removes practical coverage.

Table 5: Fixed-risk deployment over two hard splits and four budgets.

Split	Budget	Method	Coverage	Fixed success	Executed success	False-safe
combined_hard_shift	0.02	Conformal filter	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Learned risk	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Learned EU	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	CVaR recovery	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Recover-v5	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Oracle	0.14531	0.13281	0.89869	0.01250
combined_hard_shift	0.05	Conformal filter	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Learned risk	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Learned EU	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	CVaR recovery	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Recover-v5	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Oracle	0.85000	0.72187	0.84910	0.01099

Table 5: Fixed-risk deployment over two hard splits and four budgets. (continued)

Split	Budget	Method	Coverage	Fixed success	Executed success	False-safe
combined_hard_shift	0.10	Conformal filter	0.02344	0.02187	0.09333	0.00667
combined_hard_shift	0.10	Learned risk	0.33125	0.27656	0.83413	0.00857
combined_hard_shift	0.10	Learned EU	0.29063	0.24531	0.84042	0.00714
combined_hard_shift	0.10	CVaR recovery	0.02813	0.02344	0.08333	0.00000
combined_hard_shift	0.10	Recover-v5	0.45469	0.38125	0.84389	0.00400
combined_hard_shift	0.10	Oracle	0.99844	0.84844	0.84965	0.01255
combined_hard_shift	0.20	Conformal filter	0.59063	0.51719	0.87434	0.01545
combined_hard_shift	0.20	Learned risk	0.82969	0.70781	0.85326	0.01303
combined_hard_shift	0.20	Learned EU	0.79219	0.67500	0.85147	0.00835
combined_hard_shift	0.20	CVaR recovery	0.77187	0.61406	0.79331	0.01443
combined_hard_shift	0.20	Recover-v5	0.88437	0.73594	0.83242	0.00699
combined_hard_shift	0.20	Oracle	1.00000	0.85000	0.85000	0.01250
adversarial_compound_shift	0.02	Conformal filter	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.02	Learned risk	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.02	Learned EU	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.02	CVaR recovery	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.02	Recover-v5	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.02	Oracle	0.02656	0.01875	0.59286	0.00000
adversarial_compound_shift	0.05	Conformal filter	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Learned risk	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Learned EU	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	CVaR recovery	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Recover-v5	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Oracle	0.60313	0.48906	0.81022	0.00476
adversarial_compound_shift	0.10	Conformal filter	0.01094	0.00937	0.08571	0.00000
adversarial_compound_shift	0.10	Learned risk	0.26250	0.20938	0.79222	0.00556
adversarial_compound_shift	0.10	Learned EU	0.24062	0.19531	0.80935	0.00588
adversarial_compound_shift	0.10	CVaR recovery	0.01719	0.01406	0.08182	0.00000
adversarial_compound_shift	0.10	Recover-v5	0.38594	0.27969	0.72418	0.01080
adversarial_compound_shift	0.10	Oracle	0.99531	0.81250	0.81644	0.00945
adversarial_compound_shift	0.20	Conformal filter	0.51719	0.36875	0.71212	0.01649
adversarial_compound_shift	0.20	Learned risk	0.79531	0.63594	0.80148	0.01360
adversarial_compound_shift	0.20	Learned EU	0.74219	0.59219	0.80018	0.01041
adversarial_compound_shift	0.20	CVaR recovery	0.69844	0.49844	0.71291	0.01725
adversarial_compound_shift	0.20	Recover-v5	0.79688	0.60000	0.75257	0.01567
adversarial_compound_shift	0.20	Oracle	1.00000	0.81719	0.81719	0.00937

10 NEGATIVE CASES

The negative cases are not anecdotes selected to flatter the method. They include v5 failures, strong-baseline failures, ablation counterexamples, fixed-risk false-safe examples when present, and fixed-risk abstentions. They make the reviewer-facing limitation concrete: recoverability is useful only if the score can separate recoverable deviations from irreversible ones without abstaining everywhere.

Table 6: Retained negative cases.

Source	Split	Method	Plan	Outcome	Lesson
v5_main_failure	open_easy	recoverability_score_planner	direct_shortcut	unrecovered_deviation	failure was detected but recovery action did not succeed
v5_main_failure	open_easy	recoverability_score_planner	direct_shortcut	damage_or_lost_object	physical deviation damaged the object or lost manipulation state
strong_baseline_failure	open_easy	cvar_recovery_planner	direct_shortcut	irreversible_trap	nominal feasible plan entered a state with no cheap recovery branch
strong_baseline_failure	open_easy	contingent_replanner	dual_recovery_branch	unrecovered_deviation	failure was detected but recovery action did not succeed
v5_main_failure	narrow_passage_shift	recoverability_score_planner	dual_recovery_branch	budget_overrun	recovery existed but consumed too much budget
strong_baseline_failure	narrow_passage_shift	learned_expected_utility	diagnostic_staging	unrecovered_deviation	failure was detected but recovery action did not succeed
strong_baseline_failure	narrow_passage_shift	contingent_replanner	dual_recovery_branch	budget_overrun	recovery existed but consumed too much budget
strong_baseline_failure	narrow_passage_shift	learned_expected_utility	reversible_probe	budget_overrun	recovery existed but consumed too much budget
strong_baseline_failure	narrow_passage_shift	cvar_recovery_planner	dual_recovery_branch	budget_overrun	recovery existed but consumed too much budget

Table 6: Retained negative cases. (continued)

Source	Split	Method	Plan	Outcome	Lesson
strong_baseline.failure	narrow_passage.shift	cvar_recovery_planner	diagnostic_staging	budget_overrun	recovery existed but consumed too much budget
v5_main.failure	narrow_passage.shift	recoverability_score_planner	diagnostic_staging	budget_overrun	recovery existed but consumed too much budget
v5_main.failure	narrow_passage.shift	recoverability_score_planner	reversible_probe	budget_overrun	recovery existed but consumed too much budget
strong_baseline.failure	narrow_passage.shift	learned_expected_utility	dual_recovery_branch	budget_overrun	recovery existed but consumed too much budget
v5_main.failure	narrow_passage.shift	recoverability_score_planner	contingency_branch	damage_or_lost_object	physical deviation damaged the object or lost manipulation state
strong_baseline.failure	narrow_passage.shift	contingent_replanner	contingency_branch	budget_overrun	recovery existed but consumed too much budget
v5_main.failure	narrow_passage.shift	recoverability_score_planner	contingency_branch	budget_overrun	recovery existed but consumed too much budget
strong_baseline.failure	narrow_passage.shift	learned_expected_utility	diagnostic_staging	damage_or_lost_object	physical deviation damaged the object or lost manipulation state
v5_main.failure	narrow_passage.shift	recoverability_score_planner	robust_margin_route	budget_overrun	recovery existed but consumed too much budget
strong_baseline.failure	narrow_passage.shift	cvar_recovery_planner	reversible_probe	damage_or_lost_object	physical deviation damaged the object or lost manipulation state
strong_baseline.failure	narrow_passage.shift	cvar_recovery_planner	reversible_probe	budget_overrun	recovery existed but consumed too much budget
strong_baseline.failure	narrow_passage.shift	cvar_recovery_planner	contingency_branch	budget_overrun	recovery existed but consumed too much budget
v5_main.failure	narrow_passage.shift	recoverability_score_planner	diagnostic_staging	unrecovered_deviation	failure was detected but recovery action did not succeed
strong_baseline.failure	narrow_passage.shift	learned_expected_utility	diagnostic_staging	budget_overrun	recovery existed but consumed too much budget
strong_baseline.failure	narrow_passage.shift	learned_expected_utility	reversible_probe	unrecovered_deviation	failure was detected but recovery action did not succeed

11 REVIEWER ATTACK SURFACE

A hostile reviewer can make at least four fair attacks. First, learned expected utility already absorbs the same signals and wins. Second, fixed-risk abstention shows that the calibration story is not deployable yet. Third, ablations reveal that some score terms are not necessary. Fourth, the entire result remains local and generated, with no hardware or recognized high-fidelity simulator validation. The manuscript should invite these attacks rather than pretending they do not exist.

The strongest defense is modest: the repository now provides a reproducible failure analysis at meaningful scale. It separates a plausible robotics hypothesis from an unsupported submission claim. That is useful for deciding what to build next, but it is not enough for ICLR main.

12 REPRODUCIBILITY

```
python src\run_experiment.py
python scripts\generate_manuscript.py
cd paper
pdflatex -interaction=nonstopmode -halt-on-error main.tex
bibtex main
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

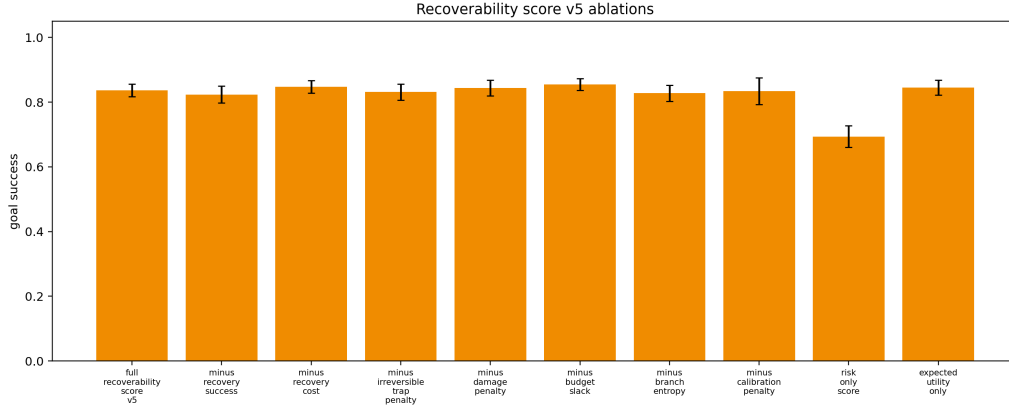


Figure 3: Combined hard-shift ablations. Central removals do not consistently degrade the method.

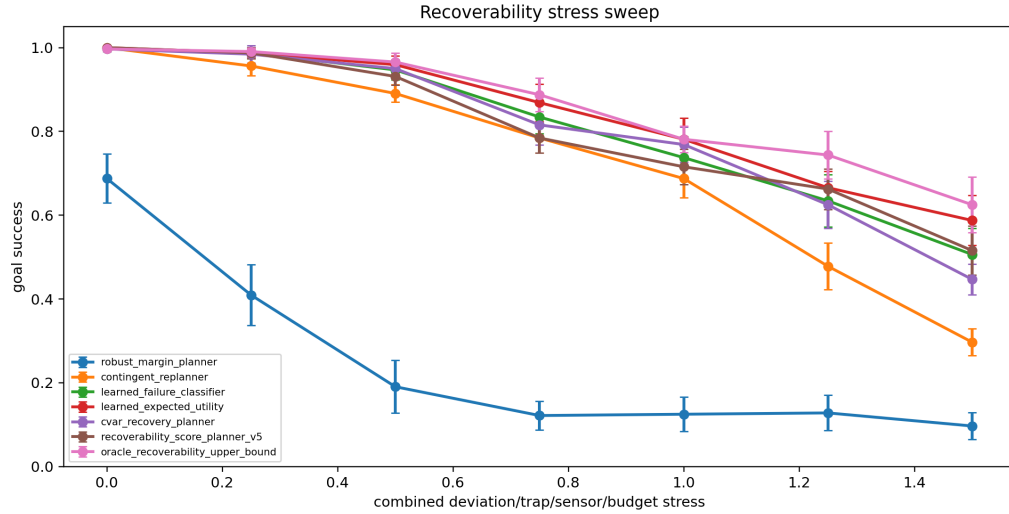


Figure 4: Combined stress sweep across seven levels. Recover-v5 does not dominate the learned expected-utility baseline at the hardest level.

```
python ../scripts/validate_submission_artifacts.py
```

The canonical numbered PDF is C:/Users/wangz/Downloads/82.pdf. No PDF should be copied to the visible Desktop. The validator checks row counts, page count, link settings, unresolved references, artifact placement, and SHA256.

13 LIMITATIONS

The limitations are not boilerplate. The local generator may not capture contact dynamics, perception failure, or planner engineering constraints of real systems. The learned baselines are still lightweight ridge models, so stronger neural or TAMP-specific baselines could widen the gap further. The oracle uses hidden probabilities and exists only as a ceiling. The fixed-risk calibration is population-bound and fails to provide useful non-oracle coverage at the strict budget. The related-work pool is broad but still needs manual expert vetting before any real submission.

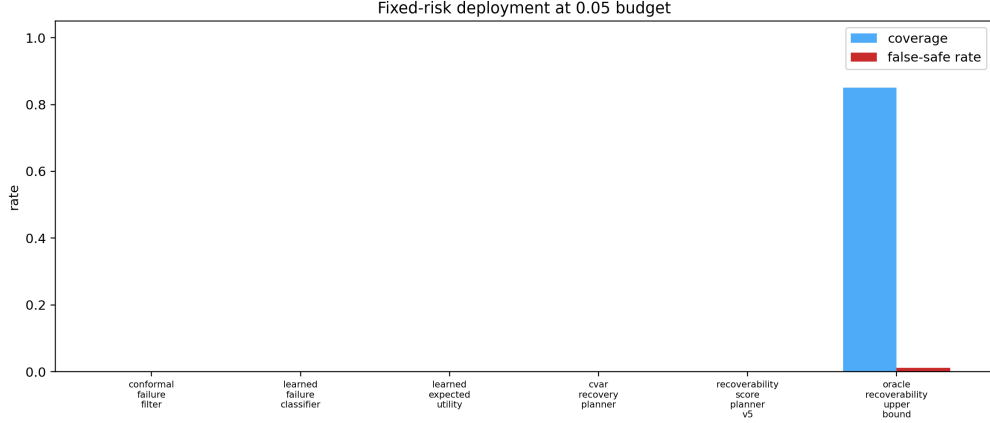


Figure 5: Fixed-risk deployment at budget 0.05 on combined hard shift. Non-oracle coverage collapses.

14 CONCLUSION

Recoverability remains a good robotics question: plans should be judged by what remains possible after the world pushes them off the nominal path. But the v5 evidence says this particular hand-designed recoverability score is not yet the answer. A learned expected-utility baseline wins the hard aggregate, ablations expose misspecification, fixed-risk coverage collapses, and maximum stress is dominated. The honest action is to archive Paper 82 as a strong negative diagnostic and move to the next paper.

A FULL SPLIT METRICS

Table 7: Per-split metrics for all methods.

Split	Method	Success	Safety utility	Irrev.	Damage
adversarial_compound_shift	Nominal	0.552 +/- 0.034	-0.433 +/- 0.117	0.2437	0.3094
adversarial_compound_shift	Risk-averse	0.575 +/- 0.028	0.226 +/- 0.048	0.0047	0.0234
adversarial_compound_shift	Robust margin	0.144 +/- 0.015	-0.441 +/- 0.035	0.0063	0.0625
adversarial_compound_shift	Expected cost	0.773 +/- 0.027	0.458 +/- 0.062	0.0172	0.0516
adversarial_compound_shift	Contingent	0.531 +/- 0.045	0.189 +/- 0.066	0.0000	0.0109
adversarial_compound_shift	RH replanner	0.733 +/- 0.033	0.461 +/- 0.048	0.0000	0.0203
adversarial_compound_shift	Conformal filter	0.750 +/- 0.027	0.493 +/- 0.047	0.0016	0.0125
adversarial_compound_shift	Learned risk	0.789 +/- 0.043	0.549 +/- 0.066	0.0000	0.0141
adversarial_compound_shift	Learned EU	0.806 +/- 0.030	0.574 +/- 0.044	0.0000	0.0141
adversarial_compound_shift	Branch entropy	0.353 +/- 0.045	-0.058 +/- 0.062	0.0000	0.0109
adversarial_compound_shift	CVaR recovery	0.722 +/- 0.029	0.433 +/- 0.048	0.0031	0.0250
adversarial_compound_shift	Recover-v5	0.734 +/- 0.037	0.459 +/- 0.064	0.0000	0.0219
adversarial_compound_shift	Oracle	0.817 +/- 0.026	0.595 +/- 0.033	0.0000	0.0094
budget_tight_shift	Nominal	0.739 +/- 0.024	0.250 +/- 0.053	0.0844	0.1453
budget_tight_shift	Risk-averse	0.531 +/- 0.034	0.195 +/- 0.048	0.0000	0.0125
budget_tight_shift	Robust margin	0.141 +/- 0.029	-0.383 +/- 0.046	0.0031	0.0234
budget_tight_shift	Expected cost	0.833 +/- 0.021	0.546 +/- 0.048	0.0219	0.0531
budget_tight_shift	Contingent	0.372 +/- 0.035	-0.012 +/- 0.049	0.0000	0.0047
budget_tight_shift	RH replanner	0.795 +/- 0.030	0.578 +/- 0.039	0.0000	0.0016
budget_tight_shift	Conformal filter	0.775 +/- 0.035	0.541 +/- 0.051	0.0000	0.0094
budget_tight_shift	Learned risk	0.786 +/- 0.028	0.565 +/- 0.039	0.0000	0.0016
budget_tight_shift	Learned EU	0.875 +/- 0.027	0.689 +/- 0.038	0.0000	0.0016
budget_tight_shift	Branch entropy	0.089 +/- 0.021	-0.408 +/- 0.031	0.0000	0.0078
budget_tight_shift	CVaR recovery	0.758 +/- 0.046	0.516 +/- 0.061	0.0000	0.0094
budget_tight_shift	Recover-v5	0.769 +/- 0.042	0.533 +/- 0.059	0.0000	0.0078
budget_tight_shift	Oracle	0.878 +/- 0.027	0.692 +/- 0.038	0.0000	0.0031
combined_hard_shift	Nominal	0.606 +/- 0.031	-0.182 +/- 0.093	0.1828	0.2344
combined_hard_shift	Risk-averse	0.698 +/- 0.031	0.395 +/- 0.053	0.0063	0.0219
combined_hard_shift	Robust margin	0.166 +/- 0.036	-0.371 +/- 0.056	0.0031	0.0375
combined_hard_shift	Expected cost	0.825 +/- 0.022	0.555 +/- 0.049	0.0172	0.0328
combined_hard_shift	Contingent	0.787 +/- 0.031	0.536 +/- 0.048	0.0000	0.0125
combined_hard_shift	RH replanner	0.827 +/- 0.030	0.601 +/- 0.047	0.0000	0.0125

Table 7: Per-split metrics for all methods. (continued)

Split	Method	Success	Safety utility	Irrev.	Damage
combined_hard_shift	Conformal filter	0.858 +/- 0.032	0.644 +/- 0.044	0.0000	0.0141
combined_hard_shift	Learned risk	0.852 +/- 0.027	0.640 +/- 0.041	0.0000	0.0109
combined_hard_shift	Learned EU	0.850 +/- 0.034	0.640 +/- 0.051	0.0000	0.0094
combined_hard_shift	Branch entropy	0.755 +/- 0.026	0.492 +/- 0.037	0.0000	0.0063
combined_hard_shift	CVaR recovery	0.809 +/- 0.034	0.572 +/- 0.054	0.0000	0.0141
combined_hard_shift	Recover-v5	0.836 +/- 0.022	0.616 +/- 0.034	0.0016	0.0063
combined_hard_shift	Oracle	0.850 +/- 0.035	0.640 +/- 0.052	0.0000	0.0125
irreversible_commitment	Nominal	0.773 +/- 0.024	0.265 +/- 0.087	0.1109	0.1469
irreversible_commitment	Risk-averse	0.833 +/- 0.015	0.608 +/- 0.024	0.0000	0.0094
irreversible_commitment	Robust margin	0.184 +/- 0.029	-0.320 +/- 0.044	0.0031	0.0219
irreversible_commitment	Expected cost	0.931 +/- 0.022	0.728 +/- 0.035	0.0141	0.0219
irreversible_commitment	Contingent	0.945 +/- 0.009	0.764 +/- 0.012	0.0000	0.0047
irreversible_commitment	RH replanner	0.983 +/- 0.008	0.835 +/- 0.013	0.0000	0.0016
irreversible_commitment	Conformal filter	0.972 +/- 0.010	0.817 +/- 0.017	0.0000	0.0047
irreversible_commitment	Learned risk	0.991 +/- 0.007	0.846 +/- 0.009	0.0000	0.0000
irreversible_commitment	Learned EU	0.980 +/- 0.012	0.835 +/- 0.014	0.0000	0.0000
irreversible_commitment	Branch entropy	0.912 +/- 0.019	0.713 +/- 0.029	0.0000	0.0016
irreversible_commitment	CVaR recovery	0.978 +/- 0.011	0.823 +/- 0.018	0.0000	0.0031
irreversible_commitment	Recover-v5	0.977 +/- 0.008	0.822 +/- 0.013	0.0000	0.0016
irreversible_commitment	Oracle	0.977 +/- 0.008	0.827 +/- 0.014	0.0000	0.0047
latent_mode_shift	Nominal	0.645 +/- 0.047	-0.007 +/- 0.139	0.1203	0.2047
latent_mode_shift	Risk-averse	0.792 +/- 0.029	0.545 +/- 0.046	0.0000	0.0141
latent_mode_shift	Robust margin	0.145 +/- 0.027	-0.413 +/- 0.036	0.0094	0.0422
latent_mode_shift	Expected cost	0.877 +/- 0.030	0.619 +/- 0.057	0.0203	0.0406
latent_mode_shift	Contingent	0.873 +/- 0.034	0.661 +/- 0.050	0.0000	0.0078
latent_mode_shift	RH replanner	0.925 +/- 0.017	0.746 +/- 0.025	0.0000	0.0094
latent_mode_shift	Conformal filter	0.917 +/- 0.025	0.737 +/- 0.039	0.0000	0.0078
latent_mode_shift	Learned risk	0.906 +/- 0.016	0.723 +/- 0.023	0.0000	0.0047
latent_mode_shift	Learned EU	0.934 +/- 0.018	0.768 +/- 0.026	0.0000	0.0016
latent_mode_shift	Branch entropy	0.816 +/- 0.037	0.579 +/- 0.049	0.0000	0.0031
latent_mode_shift	CVaR recovery	0.903 +/- 0.038	0.710 +/- 0.055	0.0000	0.0094
latent_mode_shift	Recover-v5	0.905 +/- 0.020	0.717 +/- 0.032	0.0000	0.0063
latent_mode_shift	Oracle	0.955 +/- 0.012	0.798 +/- 0.018	0.0000	0.0031
narrow_passage_shift	Nominal	0.702 +/- 0.033	0.135 +/- 0.086	0.1016	0.1766
narrow_passage_shift	Risk-averse	0.847 +/- 0.039	0.626 +/- 0.062	0.0016	0.0078
narrow_passage_shift	Robust margin	0.258 +/- 0.028	-0.211 +/- 0.042	0.0031	0.0125
narrow_passage_shift	Expected cost	0.905 +/- 0.033	0.687 +/- 0.062	0.0125	0.0266
narrow_passage_shift	Contingent	0.917 +/- 0.021	0.724 +/- 0.029	0.0000	0.0031
narrow_passage_shift	RH replanner	0.961 +/- 0.017	0.804 +/- 0.022	0.0000	0.0016
narrow_passage_shift	Conformal filter	0.975 +/- 0.015	0.823 +/- 0.023	0.0000	0.0031
narrow_passage_shift	Learned risk	0.970 +/- 0.008	0.818 +/- 0.011	0.0000	0.0000
narrow_passage_shift	Learned EU	0.973 +/- 0.011	0.821 +/- 0.017	0.0000	0.0047
narrow_passage_shift	Branch entropy	0.894 +/- 0.030	0.683 +/- 0.045	0.0000	0.0031
narrow_passage_shift	CVaR recovery	0.958 +/- 0.019	0.794 +/- 0.029	0.0000	0.0016
narrow_passage_shift	Recover-v5	0.950 +/- 0.012	0.782 +/- 0.019	0.0000	0.0031
narrow_passage_shift	Oracle	0.980 +/- 0.010	0.831 +/- 0.013	0.0000	0.0047
object_slip_shift	Nominal	0.723 +/- 0.023	0.291 +/- 0.074	0.0453	0.1391
object_slip_shift	Risk-averse	0.875 +/- 0.019	0.659 +/- 0.039	0.0016	0.0141
object_slip_shift	Robust margin	0.237 +/- 0.034	-0.248 +/- 0.055	0.0000	0.0266
object_slip_shift	Expected cost	0.866 +/- 0.023	0.584 +/- 0.052	0.0203	0.0688
object_slip_shift	Contingent	0.934 +/- 0.018	0.743 +/- 0.024	0.0000	0.0094
object_slip_shift	RH replanner	0.981 +/- 0.012	0.832 +/- 0.018	0.0000	0.0047
object_slip_shift	Conformal filter	0.980 +/- 0.010	0.828 +/- 0.018	0.0000	0.0078
object_slip_shift	Learned risk	0.981 +/- 0.013	0.827 +/- 0.024	0.0000	0.0078
object_slip_shift	Learned EU	0.991 +/- 0.007	0.846 +/- 0.012	0.0000	0.0031
object_slip_shift	Branch entropy	0.928 +/- 0.017	0.725 +/- 0.030	0.0000	0.0094
object_slip_shift	CVaR recovery	0.967 +/- 0.018	0.801 +/- 0.032	0.0000	0.0109
object_slip_shift	Recover-v5	0.967 +/- 0.013	0.807 +/- 0.021	0.0000	0.0047
object_slip_shift	Oracle	0.994 +/- 0.005	0.852 +/- 0.009	0.0000	0.0031
open_easy	Nominal	0.887 +/- 0.023	0.715 +/- 0.043	0.0047	0.0578
open_easy	Risk-averse	0.980 +/- 0.008	0.825 +/- 0.012	0.0000	0.0031
open_easy	Robust margin	0.880 +/- 0.031	0.637 +/- 0.052	0.0000	0.0141
open_easy	Expected cost	0.909 +/- 0.021	0.733 +/- 0.049	0.0063	0.0547
open_easy	Contingent	0.998 +/- 0.003	0.841 +/- 0.003	0.0000	0.0000
open_easy	RH replanner	1.000 +/- 0.000	0.863 +/- 0.000	0.0000	0.0000
open_easy	Conformal filter	1.000 +/- 0.000	0.862 +/- 0.001	0.0000	0.0000
open_easy	Learned risk	1.000 +/- 0.000	0.862 +/- 0.001	0.0000	0.0000
open_easy	Learned EU	1.000 +/- 0.000	0.864 +/- 0.000	0.0000	0.0000
open_easy	Branch entropy	1.000 +/- 0.000	0.835 +/- 0.000	0.0000	0.0000
open_easy	CVaR recovery	0.998 +/- 0.003	0.852 +/- 0.012	0.0016	0.0016
open_easy	Recover-v5	0.995 +/- 0.007	0.851 +/- 0.012	0.0000	0.0031
open_easy	Oracle	1.000 +/- 0.000	0.865 +/- 0.000	0.0000	0.0000
sensor_dropout_shift	Nominal	0.741 +/- 0.031	0.279 +/- 0.092	0.0656	0.1437
sensor_dropout_shift	Risk-averse	0.827 +/- 0.032	0.590 +/- 0.053	0.0031	0.0141
sensor_dropout_shift	Robust margin	0.202 +/- 0.037	-0.310 +/- 0.054	0.0031	0.0312
sensor_dropout_shift	Expected cost	0.889 +/- 0.013	0.638 +/- 0.032	0.0187	0.0469
sensor_dropout_shift	Contingent	0.934 +/- 0.014	0.743 +/- 0.024	0.0000	0.0109

Table 7: Per-split metrics for all methods. (continued)

Split	Method	Success	Safety utility	Irrev.	Damage
sensor_dropout_shift	RH replanner	0.975 +/- 0.010	0.818 +/- 0.019	0.0000	0.0078
sensor_dropout_shift	Conformal filter	0.972 +/- 0.011	0.818 +/- 0.017	0.0000	0.0016
sensor_dropout_shift	Learned risk	0.963 +/- 0.018	0.803 +/- 0.030	0.0000	0.0078
sensor_dropout_shift	Learned EU	0.975 +/- 0.010	0.824 +/- 0.016	0.0000	0.0063
sensor_dropout_shift	Branch entropy	0.933 +/- 0.022	0.743 +/- 0.030	0.0000	0.0000
sensor_dropout_shift	CVaR recovery	0.958 +/- 0.015	0.789 +/- 0.025	0.0000	0.0063
sensor_dropout_shift	Recover-v5	0.959 +/- 0.014	0.789 +/- 0.028	0.0016	0.0063
sensor_dropout_shift	Oracle	0.981 +/- 0.011	0.832 +/- 0.016	0.0000	0.0063

B FULL HARD-AGGREGATE PAIRWISE TABLE

The main text reports the paired comparison against the strongest non-oracle baseline. The full paired table is included here to make clear that the decision does not depend on selectively choosing one reference or one metric.

Table 8: All hard-regime aggregate paired seed differences for Recover-v5.

Reference	Metric	Mean diff	CI95	Lower95	Better seeds
Robust margin	goal_success	0.70254	0.01219	0.69035	10/10
Robust margin	recovery_success	0.28119	0.02913	0.25206	10/10
Robust margin	irreversible_failure	-0.00352	0.00149	-0.00500	9/10
Robust margin	damage	-0.02500	0.00544	-0.03044	10/10
Robust margin	total_cost	-0.53777	0.01023	-0.54800	10/10
Robust margin	calibration_error	-0.55652	0.00736	-0.56388	10/10
Robust margin	safety_utility	1.02781	0.01656	1.01125	10/10
Contingent	goal_success	0.10019	0.01011	0.09008	10/10
Contingent	recovery_success	-0.04716	0.03691	-0.08407	2/10
Contingent	irreversible_failure	0.00039	0.00051	-0.00012	0/10
Contingent	damage	-0.00078	0.00424	-0.00502	5/10
Contingent	total_cost	-0.14201	0.00894	-0.15096	10/10
Contingent	calibration_error	-0.12060	0.00854	-0.12914	10/10
Contingent	safety_utility	0.14715	0.01607	0.13108	10/10
Conformal filter	goal_success	-0.01269	0.00924	-0.02194	2/10
Conformal filter	recovery_success	0.01247	0.02929	-0.01682	6/10
Conformal filter	irreversible_failure	0.00019	0.00069	-0.00049	1/10
Conformal filter	damage	-0.00039	0.00266	-0.00306	4/10
Conformal filter	total_cost	0.04946	0.00771	0.04175	0/10
Conformal filter	calibration_error	-0.01567	0.00910	-0.02477	9/10
Conformal filter	safety_utility	-0.02200	0.01273	-0.03473	2/10
Learned risk	goal_success	-0.01758	0.01074	-0.02832	2/10
Learned risk	recovery_success	0.00535	0.02594	-0.02058	4/10
Learned risk	irreversible_failure	0.00039	0.00051	-0.00012	0/10
Learned risk	damage	0.00137	0.00243	-0.00106	3/10
Learned risk	total_cost	0.04536	0.00768	0.03768	0/10
Learned risk	calibration_error	-0.03519	0.00954	-0.04473	10/10
Learned risk	safety_utility	-0.03057	0.01614	-0.04671	2/10
Learned EU	goal_success	-0.03594	0.01288	-0.04882	0/10
Learned EU	recovery_success	0.00188	0.03609	-0.03420	4/10
Learned EU	irreversible_failure	0.00039	0.00051	-0.00012	0/10
Learned EU	damage	0.00215	0.00264	-0.00050	4/10
Learned EU	total_cost	0.07953	0.00856	0.07097	0/10
Learned EU	calibration_error	-0.02803	0.01159	-0.03962	9/10
Learned EU	safety_utility	-0.05903	0.01864	-0.07767	0/10
CVaR recovery	goal_success	0.00547	0.01330	-0.00783	5/10
CVaR recovery	recovery_success	0.02713	0.03821	-0.01107	8/10
CVaR recovery	irreversible_failure	0.00000	0.00081	-0.00081	2/10
CVaR recovery	damage	-0.00274	0.00275	-0.00548	6/10
CVaR recovery	total_cost	-0.01146	0.00933	-0.02079	8/10
CVaR recovery	calibration_error	-0.00329	0.01152	-0.01482	5/10
CVaR recovery	safety_utility	0.01090	0.02058	-0.00969	5/10

C FULL STRESS SWEEP ACROSS ALL AXES

The main text focuses on the combined stress axis because it is the hardest single diagnostic. For auditability, Table 9 reports every predefined stress axis, level, method, and aggregate metric.

Table 9: All stress axes, levels, and methods.

Axis	Level	Method	Success	CI95	Irrev.	Damage	Safety utility
navigation	0.00	Robust margin	0.21875	0.05001	0.00000	0.01562	-0.26386
navigation	0.00	Contingent	0.96875	0.00913	0.00000	0.00000	0.80081
navigation	0.00	Learned risk	0.99062	0.00936	0.00000	0.00313	0.84488
navigation	0.00	Learned EU	1.00000	0.00000	0.00000	0.00000	0.86307
navigation	0.00	CVaR recovery	0.99062	0.01838	0.00000	0.00000	0.84491
navigation	0.00	Recover-v5	0.99375	0.00817	0.00000	0.00000	0.85045
navigation	0.00	Oracle	1.00000	0.00000	0.00000	0.00000	0.86380
navigation	0.25	Robust margin	0.26875	0.06597	0.00000	0.01250	-0.18961
navigation	0.25	Contingent	0.96562	0.01429	0.00000	0.00000	0.79638
navigation	0.25	Learned risk	0.98750	0.01000	0.00000	0.00313	0.84174
navigation	0.25	Learned EU	0.99062	0.01307	0.00000	0.00000	0.84882
navigation	0.25	CVaR recovery	0.97500	0.01780	0.00000	0.00000	0.82110
navigation	0.25	Recover-v5	0.99687	0.00612	0.00000	0.00000	0.85152
navigation	0.25	Oracle	0.99687	0.00612	0.00000	0.00000	0.85965
navigation	0.50	Robust margin	0.21875	0.05001	0.00000	0.03438	-0.28795
navigation	0.50	Contingent	0.91875	0.01871	0.00000	0.00000	0.73249
navigation	0.50	Learned risk	0.97813	0.01838	0.00000	0.00000	0.83282
navigation	0.50	Learned EU	0.99062	0.01307	0.00000	0.00313	0.84576
navigation	0.50	CVaR recovery	0.95000	0.02274	0.00000	0.01250	0.77151
navigation	0.50	Recover-v5	0.97813	0.01307	0.00000	0.00313	0.82324
navigation	0.50	Oracle	0.98750	0.01000	0.00000	0.00313	0.84421
navigation	0.75	Robust margin	0.22500	0.05149	0.00313	0.02813	-0.27560
navigation	0.75	Contingent	0.93437	0.02132	0.00000	0.00313	0.75064
navigation	0.75	Learned risk	0.96562	0.02492	0.00000	0.00313	0.81096
navigation	0.75	Learned EU	0.97813	0.01838	0.00000	0.00625	0.82666
navigation	0.75	CVaR recovery	0.96875	0.01581	0.00000	0.00313	0.80585
navigation	0.75	Recover-v5	0.96875	0.01581	0.00000	0.00313	0.80819
navigation	0.75	Oracle	0.98125	0.01633	0.00000	0.00313	0.83657
navigation	1.00	Robust margin	0.20938	0.03770	0.00313	0.04375	-0.31562
navigation	1.00	Contingent	0.90625	0.02237	0.00000	0.00625	0.71053
navigation	1.00	Learned risk	0.96250	0.02000	0.00000	0.00313	0.80801
navigation	1.00	Learned EU	0.98438	0.01370	0.00313	0.00625	0.82778
navigation	1.00	CVaR recovery	0.96875	0.01826	0.00000	0.01250	0.79643
navigation	1.00	Recover-v5	0.97188	0.01099	0.00000	0.00313	0.81332
navigation	1.00	Oracle	0.98438	0.01370	0.00000	0.00313	0.83983
navigation	1.25	Robust margin	0.18438	0.04696	0.00000	0.04063	-0.34562
navigation	1.25	Contingent	0.88438	0.03298	0.00000	0.00313	0.67905
navigation	1.25	Learned risk	0.96875	0.01581	0.00000	0.00313	0.81659
navigation	1.25	Learned EU	0.98438	0.01370	0.00000	0.00625	0.83398
navigation	1.25	CVaR recovery	0.95312	0.01882	0.00000	0.01250	0.77580
navigation	1.25	Recover-v5	0.94375	0.03742	0.00000	0.00625	0.77531
navigation	1.25	Oracle	0.96875	0.01826	0.00000	0.00625	0.81974
navigation	1.50	Robust margin	0.18438	0.04870	0.00937	0.03750	-0.35296
navigation	1.50	Contingent	0.84375	0.05402	0.00000	0.00625	0.61742
navigation	1.50	Learned risk	0.95625	0.02615	0.00000	0.00625	0.79592
navigation	1.50	Learned EU	0.97188	0.02492	0.00000	0.00937	0.81267
navigation	1.50	CVaR recovery	0.94375	0.01225	0.00000	0.00937	0.76506
navigation	1.50	Recover-v5	0.94063	0.01696	0.00000	0.00625	0.76473
navigation	1.50	Oracle	0.96562	0.01926	0.00000	0.00937	0.80822
slip	0.00	Robust margin	0.29375	0.04492	0.00000	0.00625	-0.14497
slip	0.00	Contingent	0.96562	0.02132	0.00000	0.00000	0.79796
slip	0.00	Learned risk	0.99687	0.00612	0.00000	0.00000	0.85557
slip	0.00	Learned EU	0.99687	0.00612	0.00000	0.00000	0.85843
slip	0.00	CVaR recovery	0.99375	0.01225	0.00000	0.00313	0.84282
slip	0.00	Recover-v5	0.99062	0.00936	0.00000	0.00000	0.84253
slip	0.00	Oracle	0.99687	0.00612	0.00000	0.00000	0.85996
slip	0.25	Robust margin	0.22187	0.05120	0.00313	0.01562	-0.26276
slip	0.25	Contingent	0.97188	0.01429	0.00000	0.00000	0.80531
slip	0.25	Learned risk	0.99062	0.01307	0.00000	0.00000	0.84718
slip	0.25	Learned EU	0.98438	0.01021	0.00000	0.00000	0.84132
slip	0.25	CVaR recovery	0.98438	0.01370	0.00000	0.00313	0.83150
slip	0.25	Recover-v5	0.98125	0.01354	0.00000	0.00000	0.83135
slip	0.25	Oracle	0.99375	0.00817	0.00000	0.00000	0.85647
slip	0.50	Robust margin	0.23750	0.04001	0.00625	0.02813	-0.26134
slip	0.50	Contingent	0.94375	0.02381	0.00000	0.00625	0.75712
slip	0.50	Learned risk	0.99375	0.01225	0.00000	0.00000	0.85154
slip	0.50	Learned EU	0.99062	0.00936	0.00000	0.00000	0.84955
slip	0.50	CVaR recovery	0.96562	0.02319	0.00000	0.00000	0.80857
slip	0.50	Recover-v5	0.95625	0.02916	0.00000	0.00000	0.79849

Table 9: All stress axes, levels, and methods. (continued)

Axis	Level	Method	Success	CI95	Irrev.	Damage	Safety utility
slip	0.50	Oracle	0.98750	0.01354	0.00000	0.00000	0.84867
slip	0.75	Robust margin	0.29375	0.05018	0.00000	0.01250	-0.15570
slip	0.75	Contingent	0.93437	0.03222	0.00000	0.00625	0.74548
slip	0.75	Learned risk	0.98750	0.01354	0.00000	0.00313	0.83773
slip	0.75	Learned EU	0.99062	0.00936	0.00000	0.00000	0.84955
slip	0.75	CVaR recovery	0.93125	0.01780	0.00313	0.01562	0.73966
slip	0.75	Recover-v5	0.95000	0.02769	0.00000	0.00937	0.78023
slip	0.75	Oracle	0.97813	0.02246	0.00000	0.00625	0.82958
slip	1.00	Robust margin	0.22500	0.03629	0.00000	0.03438	-0.27849
slip	1.00	Contingent	0.90938	0.03090	0.00000	0.01562	0.70194
slip	1.00	Learned risk	0.98750	0.01000	0.00000	0.00313	0.83763
slip	1.00	Learned EU	0.98438	0.01370	0.00000	0.00313	0.83729
slip	1.00	CVaR recovery	0.94688	0.02747	0.00313	0.02187	0.75711
slip	1.00	Recover-v5	0.92188	0.02623	0.00000	0.03125	0.71690
slip	1.00	Oracle	0.96875	0.02042	0.00000	0.01250	0.81005
slip	1.25	Robust margin	0.24688	0.03589	0.00625	0.03438	-0.25639
slip	1.25	Contingent	0.88125	0.03136	0.00000	0.02187	0.65677
slip	1.25	Learned risk	0.98125	0.01354	0.00000	0.00937	0.82357
slip	1.25	Learned EU	0.95937	0.02246	0.00000	0.02187	0.78520
slip	1.25	CVaR recovery	0.91563	0.02246	0.00000	0.03438	0.70875
slip	1.25	Recover-v5	0.94688	0.02246	0.00000	0.01875	0.76398
slip	1.25	Oracle	0.95625	0.02082	0.00000	0.02187	0.78442
slip	1.50	Robust margin	0.21875	0.05245	0.00313	0.04688	-0.30588
slip	1.50	Contingent	0.88125	0.05229	0.00000	0.02813	0.64970
slip	1.50	Learned risk	0.97500	0.01225	0.00000	0.00625	0.81684
slip	1.50	Learned EU	0.96250	0.02709	0.00000	0.02500	0.79000
slip	1.50	CVaR recovery	0.92188	0.03681	0.00313	0.02500	0.71581
slip	1.50	Recover-v5	0.88750	0.04762	0.00000	0.02813	0.67721
slip	1.50	Oracle	0.95312	0.01882	0.00000	0.02813	0.77411
trap	0.00	Robust margin	0.16875	0.04398	0.00000	0.01250	-0.32788
trap	0.00	Contingent	0.93750	0.02416	0.00000	0.00313	0.75362
trap	0.00	Learned risk	1.00000	0.00000	0.00000	0.00000	0.85931
trap	0.00	Learned EU	0.99062	0.00936	0.00000	0.00000	0.85046
trap	0.00	CVaR recovery	0.97813	0.01307	0.00000	0.00000	0.82490
trap	0.00	Recover-v5	0.95625	0.02916	0.00000	0.00000	0.79479
trap	0.00	Oracle	0.98750	0.01354	0.00000	0.00313	0.84194
trap	0.25	Robust margin	0.15625	0.04184	0.00313	0.02187	-0.36165
trap	0.25	Contingent	0.89062	0.04938	0.00000	0.00625	0.68737
trap	0.25	Learned risk	0.97188	0.01429	0.00000	0.00000	0.82058
trap	0.25	Learned EU	0.97500	0.01780	0.00000	0.00000	0.82891
trap	0.25	CVaR recovery	0.94688	0.03298	0.00000	0.00625	0.77472
trap	0.25	Recover-v5	0.95312	0.02777	0.00000	0.00000	0.79088
trap	0.25	Oracle	0.98438	0.01021	0.00000	0.00000	0.84147
trap	0.50	Robust margin	0.16250	0.04164	0.00000	0.02187	-0.34825
trap	0.50	Contingent	0.93750	0.01291	0.00000	0.00625	0.75117
trap	0.50	Learned risk	0.97188	0.01429	0.00000	0.00000	0.81920
trap	0.50	Learned EU	0.97188	0.02654	0.00000	0.00000	0.82180
trap	0.50	CVaR recovery	0.95937	0.01838	0.00000	0.00313	0.79366
trap	0.50	Recover-v5	0.95000	0.02615	0.00313	0.00313	0.77743
trap	0.50	Oracle	0.97813	0.01307	0.00000	0.00313	0.83020
trap	0.75	Robust margin	0.17500	0.03317	0.00000	0.02187	-0.33142
trap	0.75	Contingent	0.90938	0.02807	0.00000	0.00313	0.71516
trap	0.75	Learned risk	0.97813	0.01307	0.00000	0.00000	0.82814
trap	0.75	Learned EU	0.98438	0.01370	0.00000	0.00000	0.83860
trap	0.75	CVaR recovery	0.92188	0.02923	0.00000	0.00937	0.73505
trap	0.75	Recover-v5	0.93125	0.02709	0.00000	0.00625	0.75192
trap	0.75	Oracle	0.99062	0.00936	0.00000	0.00313	0.84667
trap	1.00	Robust margin	0.17813	0.04287	0.00313	0.01562	-0.32262
trap	1.00	Contingent	0.91250	0.02381	0.00000	0.00937	0.71480
trap	1.00	Learned risk	0.98125	0.01633	0.00000	0.00000	0.83203
trap	1.00	Learned EU	0.98125	0.01871	0.00000	0.00000	0.83522
trap	1.00	CVaR recovery	0.94063	0.03090	0.00000	0.00625	0.76320
trap	1.00	Recover-v5	0.95000	0.02082	0.00000	0.00000	0.78533
trap	1.00	Oracle	0.98750	0.01000	0.00000	0.00313	0.84118
trap	1.25	Robust margin	0.14375	0.02082	0.00625	0.01875	-0.38135
trap	1.25	Contingent	0.89375	0.04302	0.00000	0.00625	0.69130
trap	1.25	Learned risk	0.96250	0.01780	0.00000	0.00000	0.80764
trap	1.25	Learned EU	0.96875	0.01826	0.00000	0.00000	0.81849
trap	1.25	CVaR recovery	0.94063	0.03222	0.00000	0.00625	0.76432
trap	1.25	Recover-v5	0.93437	0.02654	0.00000	0.00000	0.76523
trap	1.25	Oracle	0.97500	0.01225	0.00000	0.00625	0.82184
trap	1.50	Robust margin	0.15625	0.02237	0.00937	0.02187	-0.37169
trap	1.50	Contingent	0.87187	0.04129	0.00000	0.00313	0.66112
trap	1.50	Learned risk	0.95625	0.02615	0.00313	0.00313	0.78773
trap	1.50	Learned EU	0.97813	0.01307	0.00000	0.00000	0.83261
trap	1.50	CVaR recovery	0.94688	0.02424	0.00000	0.00313	0.77609
trap	1.50	Recover-v5	0.90625	0.03163	0.00000	0.00313	0.71909

Table 9: All stress axes, levels, and methods. (continued)

Axis	Level	Method	Success	CI95	Irrev.	Damage	Safety utility
trap	1.50	Oracle	0.97188	0.01696	0.00000	0.00000	0.82380
sensor	0.00	Robust margin	0.21875	0.02739	0.00313	0.00937	-0.25914
sensor	0.00	Contingent	0.91250	0.02381	0.00000	0.00313	0.71842
sensor	0.00	Learned risk	0.99375	0.00817	0.00000	0.00313	0.84665
sensor	0.00	Learned EU	0.97813	0.01307	0.00000	0.00625	0.82586
sensor	0.00	CVaR recovery	0.95937	0.01595	0.00000	0.00625	0.79304
sensor	0.00	Recover-v5	0.98125	0.01633	0.00000	0.00625	0.82226
sensor	0.00	Oracle	0.98750	0.01000	0.00000	0.00313	0.84389
sensor	0.25	Robust margin	0.22187	0.03349	0.00313	0.03125	-0.28263
sensor	0.25	Contingent	0.90312	0.02492	0.00000	0.00313	0.70445
sensor	0.25	Learned risk	0.97188	0.01099	0.00000	0.00313	0.81672
sensor	0.25	Learned EU	0.96875	0.01826	0.00000	0.01250	0.80686
sensor	0.25	CVaR recovery	0.95937	0.01838	0.00000	0.00313	0.79821
sensor	0.25	Recover-v5	0.95625	0.02082	0.00000	0.00313	0.79302
sensor	0.25	Oracle	0.97813	0.01838	0.00000	0.00313	0.83163
sensor	0.50	Robust margin	0.25312	0.05513	0.00313	0.01875	-0.22212
sensor	0.50	Contingent	0.94688	0.01837	0.00000	0.00313	0.76891
sensor	0.50	Learned risk	0.97813	0.01307	0.00000	0.00000	0.82871
sensor	0.50	Learned EU	0.97813	0.02424	0.00000	0.00937	0.82276
sensor	0.50	CVaR recovery	0.95312	0.02283	0.00000	0.01250	0.77650
sensor	0.50	Recover-v5	0.96562	0.02319	0.00000	0.01250	0.79290
sensor	0.50	Oracle	0.99062	0.00936	0.00000	0.00313	0.84843
sensor	0.75	Robust margin	0.17188	0.04307	0.00313	0.02187	-0.34029
sensor	0.75	Contingent	0.90625	0.02739	0.00000	0.01250	0.70148
sensor	0.75	Learned risk	0.97500	0.02199	0.00000	0.00313	0.82043
sensor	0.75	Learned EU	0.97500	0.01528	0.00000	0.00625	0.81997
sensor	0.75	CVaR recovery	0.97188	0.01429	0.00000	0.00625	0.80876
sensor	0.75	Recover-v5	0.94688	0.01595	0.00000	0.00625	0.77604
sensor	0.75	Oracle	0.98438	0.01370	0.00000	0.00625	0.83530
sensor	1.00	Robust margin	0.15000	0.05463	0.00313	0.02813	-0.37938
sensor	1.00	Contingent	0.91875	0.02274	0.00000	0.00313	0.73207
sensor	1.00	Learned risk	0.97500	0.01780	0.00000	0.00000	0.82400
sensor	1.00	Learned EU	0.98750	0.01633	0.00000	0.00625	0.83758
sensor	1.00	CVaR recovery	0.94375	0.01780	0.00000	0.00313	0.77165
sensor	1.00	Recover-v5	0.96562	0.01429	0.00000	0.00313	0.80441
sensor	1.00	Oracle	0.98750	0.01000	0.00000	0.00313	0.84395
sensor	1.25	Robust margin	0.26250	0.05340	0.00313	0.01875	-0.20872
sensor	1.25	Contingent	0.90938	0.03090	0.00000	0.00313	0.71975
sensor	1.25	Learned risk	0.98125	0.01354	0.00000	0.00000	0.83339
sensor	1.25	Learned EU	0.98125	0.01633	0.00000	0.00625	0.83096
sensor	1.25	CVaR recovery	0.91875	0.01354	0.00000	0.00625	0.73433
sensor	1.25	Recover-v5	0.93750	0.03028	0.00313	0.01562	0.74588
sensor	1.25	Oracle	0.97813	0.01307	0.00000	0.00313	0.83240
sensor	1.50	Robust margin	0.22500	0.05229	0.00313	0.02813	-0.27301
sensor	1.50	Contingent	0.92812	0.02895	0.00000	0.00313	0.74329
sensor	1.50	Learned risk	0.98438	0.01370	0.00000	0.00000	0.83755
sensor	1.50	Learned EU	0.97188	0.01696	0.00000	0.00625	0.81851
sensor	1.50	CVaR recovery	0.92500	0.03056	0.00000	0.00313	0.74723
sensor	1.50	Recover-v5	0.91250	0.01780	0.00000	0.01875	0.71438
sensor	1.50	Oracle	0.98750	0.01000	0.00000	0.00313	0.84431
budget	0.00	Robust margin	0.44375	0.04164	0.00313	0.02500	0.02453
budget	0.00	Contingent	0.94375	0.03513	0.00000	0.00313	0.76745
budget	0.00	Learned risk	0.98438	0.01370	0.00000	0.00000	0.83922
budget	0.00	Learned EU	0.98750	0.01354	0.00000	0.00313	0.83921
budget	0.00	CVaR recovery	0.96250	0.02381	0.00000	0.00625	0.79673
budget	0.00	Recover-v5	0.95312	0.02458	0.00000	0.01250	0.77783
budget	0.00	Oracle	0.98125	0.01633	0.00000	0.00625	0.83456
budget	0.25	Robust margin	0.23750	0.06001	0.00000	0.01562	-0.23724
budget	0.25	Contingent	0.92812	0.02591	0.00000	0.00313	0.74140
budget	0.25	Learned risk	0.95312	0.02458	0.00000	0.00000	0.79852
budget	0.25	Learned EU	0.97813	0.02052	0.00000	0.00625	0.82394
budget	0.25	CVaR recovery	0.95625	0.02082	0.00313	0.00313	0.78599
budget	0.25	Recover-v5	0.95937	0.02747	0.00000	0.00313	0.79358
budget	0.25	Oracle	0.97813	0.01595	0.00000	0.00313	0.83079
budget	0.50	Robust margin	0.17500	0.03895	0.00000	0.01250	-0.31698
budget	0.50	Contingent	0.84375	0.04744	0.00000	0.00313	0.62776
budget	0.50	Learned risk	0.92500	0.02274	0.00000	0.00313	0.75428
budget	0.50	Learned EU	0.93750	0.02042	0.00000	0.00313	0.77206
budget	0.50	CVaR recovery	0.90625	0.03536	0.00000	0.00313	0.72186
budget	0.50	Recover-v5	0.89375	0.03895	0.00313	0.00937	0.69463
budget	0.50	Oracle	0.92500	0.02916	0.00000	0.00625	0.75233
budget	0.75	Robust margin	0.14688	0.02747	0.00000	0.02500	-0.37322
budget	0.75	Contingent	0.56875	0.03742	0.00313	0.00625	0.24703
budget	0.75	Learned risk	0.88125	0.04547	0.00000	0.00313	0.69462
budget	0.75	Learned EU	0.88438	0.03879	0.00000	0.00313	0.69827
budget	0.75	CVaR recovery	0.85938	0.03901	0.00313	0.01875	0.63696
budget	0.75	Recover-v5	0.88438	0.02895	0.00000	0.00313	0.69453

Table 9: All stress axes, levels, and methods. (continued)

Axis	Level	Method	Success	CI95	Irrev.	Damage	Safety utility
budget	0.75	Oracle	0.91875	0.02274	0.00000	0.00625	0.74321
budget	1.00	Robust margin	0.12812	0.03703	0.00000	0.02813	-0.40306
budget	1.00	Contingent	0.39062	0.04678	0.00000	0.00625	0.01006
budget	1.00	Learned risk	0.78125	0.04744	0.00000	0.00625	0.55266
budget	1.00	Learned EU	0.83437	0.05558	0.00000	0.00625	0.62677
budget	1.00	CVaR recovery	0.79375	0.05181	0.00000	0.00625	0.56933
budget	1.00	Recover-v5	0.79375	0.05261	0.00000	0.01250	0.56153
budget	1.00	Oracle	0.91250	0.01780	0.00000	0.00625	0.73523
budget	1.25	Robust margin	0.14375	0.03560	0.00000	0.01875	-0.36835
budget	1.25	Contingent	0.20312	0.04006	0.00000	0.00313	-0.24046
budget	1.25	Learned risk	0.60938	0.05935	0.00000	0.00000	0.32860
budget	1.25	Learned EU	0.66563	0.06060	0.00000	0.00625	0.39778
budget	1.25	CVaR recovery	0.58750	0.06697	0.00000	0.00937	0.28676
budget	1.25	Recover-v5	0.63750	0.06966	0.00000	0.00937	0.35561
budget	1.25	Oracle	0.78438	0.05038	0.00000	0.00625	0.56168
budget	1.50	Robust margin	0.11250	0.04204	0.00000	0.03438	-0.42900
budget	1.50	Contingent	0.09688	0.02952	0.00000	0.00313	-0.38165
budget	1.50	Learned risk	0.30625	0.05973	0.00000	0.00625	-0.08737
budget	1.50	Learned EU	0.31250	0.06833	0.00000	0.00625	-0.07757
budget	1.50	CVaR recovery	0.26875	0.05789	0.00000	0.01250	-0.14565
budget	1.50	Recover-v5	0.25938	0.05328	0.00000	0.02187	-0.17015
budget	1.50	Oracle	0.35625	0.05861	0.00000	0.00313	-0.01229
combined	0.00	Robust margin	0.68750	0.05846	0.00000	0.00625	0.38623
combined	0.00	Contingent	1.00000	0.00000	0.00000	0.00000	0.84214
combined	0.00	Learned risk	0.99687	0.00612	0.00000	0.00313	0.85456
combined	0.00	Learned EU	1.00000	0.00000	0.00000	0.00000	0.86344
combined	0.00	CVaR recovery	0.99687	0.00612	0.00000	0.00313	0.84911
combined	0.00	Recover-v5	1.00000	0.00000	0.00000	0.00000	0.85856
combined	0.00	Oracle	0.99687	0.00612	0.00000	0.00000	0.86165
combined	0.25	Robust margin	0.40937	0.07273	0.00000	0.02187	-0.01608
combined	0.25	Contingent	0.95625	0.02274	0.00000	0.00313	0.78283
combined	0.25	Learned risk	0.98750	0.01354	0.00000	0.00313	0.84325
combined	0.25	Learned EU	0.98750	0.01000	0.00000	0.00000	0.84932
combined	0.25	CVaR recovery	0.98438	0.02092	0.00000	0.00000	0.83334
combined	0.25	Recover-v5	0.98750	0.01354	0.00000	0.00000	0.83989
combined	0.25	Oracle	0.99062	0.00936	0.00000	0.00313	0.84978
combined	0.50	Robust margin	0.19062	0.06290	0.00625	0.03125	-0.33485
combined	0.50	Contingent	0.89062	0.02092	0.00000	0.01250	0.67875
combined	0.50	Learned risk	0.94688	0.02424	0.00000	0.00937	0.77981
combined	0.50	Learned EU	0.95937	0.02052	0.00000	0.00625	0.80205
combined	0.50	CVaR recovery	0.95000	0.02082	0.00000	0.00313	0.77880
combined	0.50	Recover-v5	0.93125	0.02000	0.00313	0.00625	0.75033
combined	0.50	Oracle	0.96562	0.02132	0.00000	0.00937	0.80627
combined	0.75	Robust margin	0.12187	0.03471	0.00625	0.05312	-0.46050
combined	0.75	Contingent	0.78438	0.03589	0.00000	0.02187	0.52107
combined	0.75	Learned risk	0.83437	0.03985	0.00000	0.02500	0.60058
combined	0.75	Learned EU	0.86875	0.04454	0.00000	0.00937	0.66589
combined	0.75	CVaR recovery	0.81563	0.04784	0.00000	0.01875	0.57360
combined	0.75	Recover-v5	0.78438	0.03589	0.00000	0.00937	0.54397
combined	0.75	Oracle	0.88750	0.04001	0.00000	0.01250	0.69522
combined	1.00	Robust margin	0.12500	0.04083	0.02813	0.06250	-0.50444
combined	1.00	Contingent	0.68750	0.04565	0.00000	0.02187	0.38334
combined	1.00	Learned risk	0.73750	0.05018	0.00000	0.03125	0.45138
combined	1.00	Learned EU	0.78125	0.05084	0.00000	0.00937	0.54207
combined	1.00	CVaR recovery	0.76875	0.04204	0.00000	0.01250	0.51457
combined	1.00	Recover-v5	0.71562	0.04229	0.00000	0.03125	0.41746
combined	1.00	Oracle	0.78125	0.03163	0.00000	0.01562	0.53586
combined	1.25	Robust margin	0.12812	0.04229	0.02500	0.06563	-0.50471
combined	1.25	Contingent	0.47813	0.05558	0.00000	0.05000	0.05904
combined	1.25	Learned risk	0.63438	0.06263	0.00000	0.03438	0.30335
combined	1.25	Learned EU	0.66563	0.03879	0.00000	0.03125	0.35230
combined	1.25	CVaR recovery	0.62500	0.05628	0.00313	0.04688	0.26775
combined	1.25	Recover-v5	0.66250	0.04814	0.00313	0.02813	0.34359
combined	1.25	Oracle	0.74375	0.05687	0.00000	0.02500	0.46962
combined	1.50	Robust margin	0.09688	0.03222	0.05625	0.13125	-0.67746
combined	1.50	Contingent	0.29688	0.03196	0.00000	0.05937	-0.20394
combined	1.50	Learned risk	0.50625	0.06246	0.00313	0.05312	0.09506
combined	1.50	Learned EU	0.58750	0.05973	0.00000	0.03750	0.23198
combined	1.50	CVaR recovery	0.44688	0.03658	0.01250	0.03125	0.02236
combined	1.50	Recover-v5	0.51562	0.05864	0.00625	0.03750	0.12029
combined	1.50	Oracle	0.62500	0.06584	0.00000	0.04688	0.27470

D ABLATION SEED-LEVEL METRICS

The ablation gate fails at the aggregate level, but seed-level rows are retained so that the failure cannot be dismissed as a table-generation artifact. These rows are generated directly from `ablation_seedmetrics.csv`.

Table 10: Seed-level ablation metrics.

Split	Ablation	Seed	Episodes	Success	Irrev.	Damage	Safety utility
	adversarial_compound_shiftfull_recoverability_score_v5	0	80	0.72500	0.00000	0.01250	0.45859
	adversarial_compound_shiftfull_recoverability_score_v5	1	80	0.72500	0.00000	0.00000	0.47130
	adversarial_compound_shiftfull_recoverability_score_v5	2	80	0.72500	0.00000	0.00000	0.47206
	adversarial_compound_shiftfull_recoverability_score_v5	3	80	0.66250	0.00000	0.06250	0.30877
	adversarial_compound_shiftfull_recoverability_score_v5	4	80	0.71250	0.00000	0.02500	0.42336
	adversarial_compound_shiftfull_recoverability_score_v5	5	80	0.73750	0.00000	0.01250	0.47554
	adversarial_compound_shiftfull_recoverability_score_v5	6	80	0.76250	0.00000	0.02500	0.49549
	adversarial_compound_shiftfull_recoverability_score_v5	7	80	0.81250	0.00000	0.01250	0.58332
	adversarial_compound_shiftfull_recoverability_score_v5	8	80	0.68750	0.00000	0.03750	0.37322
	adversarial_compound_shiftfull_recoverability_score_v5	9	80	0.72500	0.00000	0.02500	0.44390
	adversarial_compound_shiftminus_recovery_success	0	80	0.80000	0.00000	0.01250	0.56311
	adversarial_compound_shiftminus_recovery_success	1	80	0.67500	0.00000	0.00000	0.40309
	adversarial_compound_shiftminus_recovery_success	2	80	0.73750	0.00000	0.05000	0.42985
	adversarial_compound_shiftminus_recovery_success	3	80	0.71250	0.01250	0.01250	0.42435
	adversarial_compound_shiftminus_recovery_success	4	80	0.67500	0.02500	0.02500	0.34014
	adversarial_compound_shiftminus_recovery_success	5	80	0.65000	0.00000	0.00000	0.36515
	adversarial_compound_shiftminus_recovery_success	6	80	0.80000	0.01250	0.01250	0.54096
	adversarial_compound_shiftminus_recovery_success	7	80	0.73750	0.00000	0.00000	0.48876
	adversarial_compound_shiftminus_recovery_success	8	80	0.65000	0.00000	0.02500	0.33625
	adversarial_compound_shiftminus_recovery_success	9	80	0.71250	0.00000	0.01250	0.44083
	adversarial_compound_shiftminus_recovery_cost	0	80	0.81250	0.00000	0.05000	0.53557
	adversarial_compound_shiftminus_recovery_cost	1	80	0.62500	0.00000	0.02500	0.29995
	adversarial_compound_shiftminus_recovery_cost	2	80	0.76250	0.00000	0.02500	0.49423
	adversarial_compound_shiftminus_recovery_cost	3	80	0.70000	0.00000	0.02500	0.40997
	adversarial_compound_shiftminus_recovery_cost	4	80	0.68750	0.00000	0.02500	0.38875
	adversarial_compound_shiftminus_recovery_cost	5	80	0.70000	0.00000	0.02500	0.40841
	adversarial_compound_shiftminus_recovery_cost	6	80	0.73750	0.00000	0.03750	0.44486
	adversarial_compound_shiftminus_recovery_cost	7	80	0.66250	0.00000	0.05000	0.32458
	adversarial_compound_shiftminus_recovery_cost	8	80	0.76250	0.01250	0.02500	0.47719
	adversarial_compound_shiftminus_recovery_cost	9	80	0.72500	0.00000	0.01250	0.45837
	adversarial_compound_shiftminus_irreversible_trap_penalty	0	80	0.81250	0.00000	0.02500	0.56745
	adversarial_compound_shiftminus_irreversible_trap_penalty	1	80	0.71250	0.00000	0.00000	0.45488
	adversarial_compound_shiftminus_irreversible_trap_penalty	2	80	0.75000	0.00000	0.01250	0.49204
	adversarial_compound_shiftminus_irreversible_trap_penalty	3	80	0.72500	0.00000	0.03750	0.42625
	adversarial_compound_shiftminus_irreversible_trap_penalty	4	80	0.81250	0.00000	0.02500	0.56438
	adversarial_compound_shiftminus_irreversible_trap_penalty	5	80	0.67500	0.00000	0.01250	0.38715
	adversarial_compound_shiftminus_irreversible_trap_penalty	6	80	0.66250	0.00000	0.03750	0.33953
	adversarial_compound_shiftminus_irreversible_trap_penalty	7	80	0.78750	0.00000	0.01250	0.54316
	adversarial_compound_shiftminus_irreversible_trap_penalty	8	80	0.68750	0.00000	0.00000	0.41981
	adversarial_compound_shiftminus_irreversible_trap_penalty	9	80	0.71250	0.00000	0.00000	0.45600
	adversarial_compound_shiftminus_damage_penalty	0	80	0.82500	0.00000	0.01250	0.59982
	adversarial_compound_shiftminus_damage_penalty	1	80	0.72500	0.00000	0.01250	0.45771
	adversarial_compound_shiftminus_damage_penalty	2	80	0.71250	0.00000	0.01250	0.44176
	adversarial_compound_shiftminus_damage_penalty	3	80	0.75000	0.01250	0.02500	0.46014
	adversarial_compound_shiftminus_damage_penalty	4	80	0.77500	0.00000	0.01250	0.52736
	adversarial_compound_shiftminus_damage_penalty	5	80	0.80000	0.00000	0.02500	0.54774
	adversarial_compound_shiftminus_damage_penalty	6	80	0.71250	0.00000	0.01250	0.44154
	adversarial_compound_shiftminus_damage_penalty	7	80	0.82500	0.00000	0.00000	0.61218
	adversarial_compound_shiftminus_damage_penalty	8	80	0.73750	0.00000	0.00000	0.48978
	adversarial_compound_shiftminus_damage_penalty	9	80	0.72500	0.00000	0.03750	0.42672
	adversarial_compound_shiftminus_budget_slack	0	80	0.70000	0.00000	0.02500	0.41065
	adversarial_compound_shiftminus_budget_slack	1	80	0.72500	0.00000	0.02500	0.44084
	adversarial_compound_shiftminus_budget_slack	2	80	0.66250	0.00000	0.01250	0.37157
	adversarial_compound_shiftminus_budget_slack	3	80	0.58750	0.00000	0.02500	0.24971
	adversarial_compound_shiftminus_budget_slack	4	80	0.65000	0.01250	0.01250	0.33718
	adversarial_compound_shiftminus_budget_slack	5	80	0.76250	0.00000	0.02500	0.49536
	adversarial_compound_shiftminus_budget_slack	6	80	0.66250	0.00000	0.01250	0.37152
	adversarial_compound_shiftminus_budget_slack	7	80	0.73750	0.00000	0.02500	0.45946
	adversarial_compound_shiftminus_budget_slack	8	80	0.71250	0.00000	0.02500	0.42337
	adversarial_compound_shiftminus_budget_slack	9	80	0.65000	0.00000	0.01250	0.35508
	adversarial_compound_shiftminus_branch_entropy	0	80	0.72500	0.00000	0.00000	0.47397
	adversarial_compound_shiftminus_branch_entropy	1	80	0.75000	0.00000	0.00000	0.50593
	adversarial_compound_shiftminus_branch_entropy	2	80	0.82500	0.00000	0.00000	0.61359
	adversarial_compound_shiftminus_branch_entropy	3	80	0.70000	0.00000	0.03750	0.39249
	adversarial_compound_shiftminus_branch_entropy	4	80	0.73750	0.00000	0.00000	0.49098
	adversarial_compound_shiftminus_branch_entropy	5	80	0.77500	0.00000	0.01250	0.52738
	adversarial_compound_shiftminus_branch_entropy	6	80	0.78750	0.00000	0.01250	0.54563
	adversarial_compound_shiftminus_branch_entropy	7	80	0.72500	0.00000	0.00000	0.47151

Table 10: Seed-level ablation metrics. (continued)

Split	Ablation	Seed	Episodes	Success	Irrev.	Damage	Safety utility
adversarial_compound_shiftminus_branch_entropy		8	80	0.68750	0.00000	0.01250	0.40359
adversarial_compound_shiftminus_branch_entropy		9	80	0.65000	0.00000	0.02500	0.33719
adversarial_compound_shiftminus_calibration_penalty		0	80	0.75000	0.00000	0.00000	0.50875
adversarial_compound_shiftminus_calibration_penalty		1	80	0.72500	0.00000	0.02500	0.44183
adversarial_compound_shiftminus_calibration_penalty		2	80	0.68750	0.00000	0.01250	0.40580
adversarial_compound_shiftminus_calibration_penalty		3	80	0.77500	0.00000	0.01250	0.52964
adversarial_compound_shiftminus_calibration_penalty		4	80	0.70000	0.00000	0.02500	0.40711
adversarial_compound_shiftminus_calibration_penalty		5	80	0.75000	0.00000	0.02500	0.47898
adversarial_compound_shiftminus_calibration_penalty		6	80	0.75000	0.00000	0.02500	0.47863
adversarial_compound_shiftminus_calibration_penalty		7	80	0.76250	0.00000	0.02500	0.49419
adversarial_compound_shiftminus_calibration_penalty		8	80	0.66250	0.00000	0.03750	0.34000
adversarial_compound_shiftminus_calibration_penalty		9	80	0.75000	0.00000	0.00000	0.50804
adversarial_compound_shiftrisk_only_score		0	80	0.58750	0.00000	0.01250	0.26558
adversarial_compound_shiftrisk_only_score		1	80	0.48750	0.00000	0.03750	0.09104
adversarial_compound_shiftrisk_only_score		2	80	0.57500	0.01250	0.05000	0.18519
adversarial_compound_shiftrisk_only_score		3	80	0.50000	0.01250	0.05000	0.07711
adversarial_compound_shiftrisk_only_score		4	80	0.52500	0.00000	0.02500	0.16332
adversarial_compound_shiftrisk_only_score		5	80	0.55000	0.00000	0.02500	0.19620
adversarial_compound_shiftrisk_only_score		6	80	0.55000	0.01250	0.01250	0.19216
adversarial_compound_shiftrisk_only_score		7	80	0.57500	0.00000	0.03750	0.21569
adversarial_compound_shiftrisk_only_score		8	80	0.62500	0.00000	0.00000	0.33259
adversarial_compound_shiftrisk_only_score		9	80	0.52500	0.00000	0.01250	0.17632
adversarial_compound_shiftexpected_utility_only		0	80	0.78750	0.00000	0.01250	0.54835
adversarial_compound_shiftexpected_utility_only		1	80	0.73750	0.00000	0.00000	0.48927
adversarial_compound_shiftexpected_utility_only		2	80	0.77500	0.01250	0.03750	0.48149
adversarial_compound_shiftexpected_utility_only		3	80	0.76250	0.00000	0.02500	0.49450
adversarial_compound_shiftexpected_utility_only		4	80	0.73750	0.01250	0.05000	0.41338
adversarial_compound_shiftexpected_utility_only		5	80	0.77500	0.00000	0.01250	0.52795
adversarial_compound_shiftexpected_utility_only		6	80	0.78750	0.00000	0.01250	0.54577
adversarial_compound_shiftexpected_utility_only		7	80	0.76250	0.00000	0.01250	0.50846
adversarial_compound_shiftexpected_utility_only		8	80	0.70000	0.00000	0.01250	0.42075
adversarial_compound_shiftexpected_utility_only		9	80	0.81250	0.00000	0.00000	0.59599
combined_hard_shift_full_recoverability_score_v5		0	80	0.83750	0.00000	0.00000	0.62893
combined_hard_shift_full_recoverability_score_v5		1	80	0.86250	0.00000	0.00000	0.66290
combined_hard_shift_full_recoverability_score_v5		2	80	0.88750	0.00000	0.02500	0.66746
combined_hard_shift_full_recoverability_score_v5		3	80	0.86250	0.00000	0.02500	0.63223
combined_hard_shift_full_recoverability_score_v5		4	80	0.83750	0.00000	0.00000	0.62884
combined_hard_shift_full_recoverability_score_v5		5	80	0.83750	0.00000	0.00000	0.62880
combined_hard_shift_full_recoverability_score_v5		6	80	0.77500	0.00000	0.00000	0.53842
combined_hard_shift_full_recoverability_score_v5		7	80	0.81250	0.00000	0.00000	0.59205
combined_hard_shift_full_recoverability_score_v5		8	80	0.83750	0.00000	0.01250	0.61421
combined_hard_shift_full_recoverability_score_v5		9	80	0.81250	0.01250	0.03750	0.53375
combined_hard_shift_minus_recovery_success		0	80	0.87500	0.00000	0.01250	0.66949
combined_hard_shift_minus_recovery_success		1	80	0.80000	0.00000	0.02500	0.54953
combined_hard_shift_minus_recovery_success		2	80	0.77500	0.00000	0.02500	0.50987
combined_hard_shift_minus_recovery_success		3	80	0.81250	0.00000	0.03750	0.54720
combined_hard_shift_minus_recovery_success		4	80	0.85000	0.00000	0.00000	0.64548
combined_hard_shift_minus_recovery_success		5	80	0.75000	0.01250	0.05000	0.43491
combined_hard_shift_minus_recovery_success		6	80	0.81250	0.00000	0.01250	0.57568
combined_hard_shift_minus_recovery_success		7	80	0.82500	0.00000	0.01250	0.59492
combined_hard_shift_minus_recovery_success		8	80	0.85000	0.00000	0.02500	0.61552
combined_hard_shift_minus_recovery_success		9	80	0.88750	0.00000	0.00000	0.69701
combined_hard_shift_minus_recovery_cost		0	80	0.85000	0.00000	0.01250	0.63109
combined_hard_shift_minus_recovery_cost		1	80	0.85000	0.00000	0.03750	0.59857
combined_hard_shift_minus_recovery_cost		2	80	0.78750	0.00000	0.05000	0.49662
combined_hard_shift_minus_recovery_cost		3	80	0.88750	0.00000	0.02500	0.66723
combined_hard_shift_minus_recovery_cost		4	80	0.87500	0.00000	0.01250	0.66452
combined_hard_shift_minus_recovery_cost		5	80	0.85000	0.00000	0.01250	0.63098
combined_hard_shift_minus_recovery_cost		6	80	0.82500	0.00000	0.00000	0.60897
combined_hard_shift_minus_recovery_cost		7	80	0.87500	0.00000	0.00000	0.67931
combined_hard_shift_minus_recovery_cost		8	80	0.86250	0.00000	0.03750	0.62441
combined_hard_shift_minus_recovery_cost		9	80	0.81250	0.00000	0.06250	0.51584
combined_hard_shift_minus_irreversible_trap_penalty		0	80	0.83750	0.00000	0.01250	0.61500
combined_hard_shift_minus_irreversible_trap_penalty		1	80	0.87500	0.00000	0.01250	0.66830
combined_hard_shift_minus_irreversible_trap_penalty		2	80	0.85000	0.00000	0.02500	0.61667
combined_hard_shift_minus_irreversible_trap_penalty		3	80	0.86250	0.00000	0.01250	0.65229
combined_hard_shift_minus_irreversible_trap_penalty		4	80	0.80000	0.00000	0.01250	0.56209
combined_hard_shift_minus_irreversible_trap_penalty		5	80	0.73750	0.00000	0.00000	0.49019
combined_hard_shift_minus_irreversible_trap_penalty		6	80	0.82500	0.00000	0.01250	0.59510
combined_hard_shift_minus_irreversible_trap_penalty		7	80	0.86250	0.00000	0.00000	0.66229
combined_hard_shift_minus_irreversible_trap_penalty		8	80	0.82500	0.00000	0.00000	0.61211
combined_hard_shift_minus_irreversible_trap_penalty		9	80	0.83750	0.00000	0.00000	0.62705
combined_hard_shift_minus_damage_penalty		0	80	0.88750	0.00000	0.01250	0.68835
combined_hard_shift_minus_damage_penalty		1	80	0.85000	0.00000	0.00000	0.64478
combined_hard_shift_minus_damage_penalty		2	80	0.77500	0.00000	0.01250	0.52720
combined_hard_shift_minus_damage_penalty		3	80	0.81250	0.00000	0.02500	0.56351
combined_hard_shift_minus_damage_penalty		4	80	0.81250	0.00000	0.00000	0.59534

Table 10: Seed-level ablation metrics. (continued)

Split	Ablation	Seed	Episodes	Success	Irrev.	Damage	Safety utility
combined_hard_shift	minus_damage_penalty	5	80	0.86250	0.00000	0.02500	0.63457
combined_hard_shift	minus_damage_penalty	6	80	0.81250	0.00000	0.01250	0.57505
combined_hard_shift	minus_damage_penalty	7	80	0.88750	0.00000	0.02500	0.66720
combined_hard_shift	minus_damage_penalty	8	80	0.88750	0.00000	0.03750	0.65298
combined_hard_shift	minus_damage_penalty	9	80	0.85000	0.00000	0.02500	0.61570
combined_hard_shift	minus_budget_slack	0	80	0.83750	0.00000	0.01250	0.61297
combined_hard_shift	minus_budget_slack	1	80	0.80000	0.00000	0.00000	0.57467
combined_hard_shift	minus_budget_slack	2	80	0.88750	0.00000	0.01250	0.68178
combined_hard_shift	minus_budget_slack	3	80	0.87500	0.00000	0.01250	0.66247
combined_hard_shift	minus_budget_slack	4	80	0.90000	0.00000	0.00000	0.71904
combined_hard_shift	minus_budget_slack	5	80	0.82500	0.00000	0.01250	0.59668
combined_hard_shift	minus_budget_slack	6	80	0.86250	0.00000	0.02500	0.63148
combined_hard_shift	minus_budget_slack	7	80	0.86250	0.00000	0.00000	0.66196
combined_hard_shift	minus_budget_slack	8	80	0.85000	0.00000	0.01250	0.62958
combined_hard_shift	minus_budget_slack	9	80	0.85000	0.00000	0.00000	0.64503
combined_hard_shift	minus_branch_entropy	0	80	0.87500	0.00000	0.01250	0.66746
combined_hard_shift	minus_branch_entropy	1	80	0.83750	0.00000	0.01250	0.61431
combined_hard_shift	minus_branch_entropy	2	80	0.80000	0.00000	0.00000	0.57575
combined_hard_shift	minus_branch_entropy	3	80	0.75000	0.00000	0.00000	0.50547
combined_hard_shift	minus_branch_entropy	4	80	0.85000	0.00000	0.01250	0.63213
combined_hard_shift	minus_branch_entropy	5	80	0.87500	0.00000	0.01250	0.66806
combined_hard_shift	minus_branch_entropy	6	80	0.80000	0.00000	0.01250	0.55893
combined_hard_shift	minus_branch_entropy	7	80	0.85000	0.00000	0.01250	0.63102
combined_hard_shift	minus_branch_entropy	8	80	0.83750	0.00000	0.01250	0.61532
combined_hard_shift	minus_branch_entropy	9	80	0.80000	0.00000	0.02500	0.54514
combined_hard_shift	minus_calibration_penalty	0	80	0.86250	0.00000	0.05000	0.60355
combined_hard_shift	minus_calibration_penalty	1	80	0.85000	0.00000	0.00000	0.64474
combined_hard_shift	minus_calibration_penalty	2	80	0.77500	0.00000	0.02500	0.51070
combined_hard_shift	minus_calibration_penalty	3	80	0.83750	0.00000	0.01250	0.61233
combined_hard_shift	minus_calibration_penalty	4	80	0.87500	0.00000	0.01250	0.66646
combined_hard_shift	minus_calibration_penalty	5	80	0.70000	0.00000	0.03750	0.39181
combined_hard_shift	minus_calibration_penalty	6	80	0.82500	0.00000	0.01250	0.59390
combined_hard_shift	minus_calibration_penalty	7	80	0.78750	0.00000	0.00000	0.55635
combined_hard_shift	minus_calibration_penalty	8	80	0.88750	0.00000	0.00000	0.69940
combined_hard_shift	minus_calibration_penalty	9	80	0.93750	0.00000	0.00000	0.77155
combined_hard_shift	risk_only_score	0	80	0.68750	0.00000	0.00000	0.41577
combined_hard_shift	risk_only_score	1	80	0.66250	0.00000	0.02500	0.34808
combined_hard_shift	risk_only_score	2	80	0.60000	0.00000	0.02500	0.26343
combined_hard_shift	risk_only_score	3	80	0.81250	0.00000	0.02500	0.55941
combined_hard_shift	risk_only_score	4	80	0.71250	0.01250	0.01250	0.41895
combined_hard_shift	risk_only_score	5	80	0.68750	0.00000	0.00000	0.41631
combined_hard_shift	risk_only_score	6	80	0.72500	0.01250	0.02500	0.41900
combined_hard_shift	risk_only_score	7	80	0.67500	0.00000	0.03750	0.34519
combined_hard_shift	risk_only_score	8	80	0.67500	0.00000	0.01250	0.38087
combined_hard_shift	risk_only_score	9	80	0.70000	0.00000	0.05000	0.37135
combined_hard_shift	expected_utility_only	0	80	0.85000	0.00000	0.02500	0.61663
combined_hard_shift	expected_utility_only	1	80	0.88750	0.00000	0.00000	0.69983
combined_hard_shift	expected_utility_only	2	80	0.91250	0.00000	0.00000	0.73820
combined_hard_shift	expected_utility_only	3	80	0.80000	0.00000	0.02500	0.54398
combined_hard_shift	expected_utility_only	4	80	0.86250	0.01250	0.02500	0.61905
combined_hard_shift	expected_utility_only	5	80	0.80000	0.00000	0.03750	0.53622
combined_hard_shift	expected_utility_only	6	80	0.82500	0.00000	0.00000	0.60999
combined_hard_shift	expected_utility_only	7	80	0.82500	0.00000	0.00000	0.60907
combined_hard_shift	expected_utility_only	8	80	0.86250	0.00000	0.00000	0.66560
combined_hard_shift	expected_utility_only	9	80	0.82500	0.00000	0.00000	0.60981

E FIXED-RISK SEED-LEVEL METRICS

Fixed-risk coverage is the most submission-damaging result: non-oracle methods abstain at the strict 0.05 budget. The seed-level table below shows this is consistent across seeds rather than an averaging artifact.

Table 11: Seed-level fixed-risk metrics.

Split	Budget	Method	Seed	Coverage	Fixed success	Executed success	False-safe
combined_hard_shift	0.02	Conformal filter	0	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Conformal filter	1	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.02	Conformal filter	2	0.00000	0.00000	0.00000	0.00000

Table 11: Seed-level fixed-risk metrics. (continued)

[illegible]

Table 11: Seed-level fixed-risk metrics. (continued)

Split	Budget	Method	Seed	Coverage	Fixed success	Executed success	False-safe
combined_hard_shift	0.05	Learned EU	0	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Learned EU	1	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Learned EU	2	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Learned EU	3	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Learned EU	4	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Learned EU	5	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Learned EU	6	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Learned EU	7	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Learned EU	8	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Learned EU	9	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	CVaR recovery	0	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	CVaR recovery	1	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	CVaR recovery	2	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	CVaR recovery	3	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	CVaR recovery	4	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	CVaR recovery	5	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	CVaR recovery	6	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	CVaR recovery	7	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	CVaR recovery	8	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	CVaR recovery	9	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Recover-v5	0	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Recover-v5	1	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Recover-v5	2	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Recover-v5	3	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Recover-v5	4	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Recover-v5	5	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Recover-v5	6	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Recover-v5	7	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Recover-v5	8	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Recover-v5	9	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.05	Oracle	0	0.84375	0.70312	0.83333	0.00000
combined_hard_shift	0.05	Oracle	1	0.87500	0.76562	0.87500	0.01786
combined_hard_shift	0.05	Oracle	2	0.81250	0.68750	0.84615	0.00000
combined_hard_shift	0.05	Oracle	3	0.78125	0.73438	0.94000	0.00000
combined_hard_shift	0.05	Oracle	4	0.87500	0.76562	0.87500	0.01786
combined_hard_shift	0.05	Oracle	5	0.81250	0.64062	0.78846	0.01923
combined_hard_shift	0.05	Oracle	6	0.82812	0.64062	0.77358	0.01887
combined_hard_shift	0.05	Oracle	7	0.90625	0.79688	0.87931	0.01724
combined_hard_shift	0.05	Oracle	8	0.82812	0.68750	0.83019	0.01887
combined_hard_shift	0.05	Oracle	9	0.93750	0.79688	0.85000	0.00000
combined_hard_shift	0.10	Conformal filter	0	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Conformal filter	1	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Conformal filter	2	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Conformal filter	3	0.23438	0.21875	0.93333	0.06667
combined_hard_shift	0.10	Conformal filter	4	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Conformal filter	5	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Conformal filter	6	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Conformal filter	7	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Conformal filter	8	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Conformal filter	9	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Learned risk	0	0.29688	0.25000	0.84211	0.00000
combined_hard_shift	0.10	Learned risk	1	0.35938	0.28125	0.78261	0.00000
combined_hard_shift	0.10	Learned risk	2	0.31250	0.23438	0.75000	0.00000
combined_hard_shift	0.10	Learned risk	3	0.28125	0.23438	0.83333	0.00000
combined_hard_shift	0.10	Learned risk	4	0.34375	0.28125	0.81818	0.00000
combined_hard_shift	0.10	Learned risk	5	0.31250	0.26562	0.85000	0.05000
combined_hard_shift	0.10	Learned risk	6	0.42188	0.37500	0.88889	0.00000
combined_hard_shift	0.10	Learned risk	7	0.23438	0.21875	0.93333	0.00000
combined_hard_shift	0.10	Learned risk	8	0.31250	0.23438	0.75000	0.00000
combined_hard_shift	0.10	Learned risk	9	0.43750	0.39062	0.89286	0.03571
combined_hard_shift	0.10	Learned EU	0	0.26562	0.25000	0.94118	0.00000
combined_hard_shift	0.10	Learned EU	1	0.28125	0.28125	1.00000	0.00000
combined_hard_shift	0.10	Learned EU	2	0.25000	0.20312	0.81250	0.00000
combined_hard_shift	0.10	Learned EU	3	0.29688	0.21875	0.73684	0.00000
combined_hard_shift	0.10	Learned EU	4	0.29688	0.25000	0.84211	0.00000
combined_hard_shift	0.10	Learned EU	5	0.28125	0.23438	0.83333	0.00000
combined_hard_shift	0.10	Learned EU	6	0.39062	0.31250	0.80000	0.00000
combined_hard_shift	0.10	Learned EU	7	0.21875	0.14062	0.64286	0.07143
combined_hard_shift	0.10	Learned EU	8	0.26562	0.23438	0.88235	0.00000
combined_hard_shift	0.10	Learned EU	9	0.35938	0.32812	0.91304	0.00000
combined_hard_shift	0.10	CVaR recovery	0	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	CVaR recovery	1	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	CVaR recovery	2	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	CVaR recovery	3	0.28125	0.23438	0.83333	0.00000
combined_hard_shift	0.10	CVaR recovery	4	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	CVaR recovery	5	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	CVaR recovery	6	0.00000	0.00000	0.00000	0.00000

Table 11: Seed-level fixed-risk metrics. (continued)

Split	Budget	Method	Seed	Coverage	Fixed success	Executed success	False-safe
combined_hard_shift	0.10	CVaR recovery	7	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	CVaR recovery	8	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	CVaR recovery	9	0.00000	0.00000	0.00000	0.00000
combined_hard_shift	0.10	Recover-v5	0	0.37500	0.32812	0.87500	0.00000
combined_hard_shift	0.10	Recover-v5	1	0.40625	0.35938	0.88462	0.00000
combined_hard_shift	0.10	Recover-v5	2	0.39062	0.35938	0.92000	0.04000
combined_hard_shift	0.10	Recover-v5	3	0.51562	0.43750	0.84848	0.00000
combined_hard_shift	0.10	Recover-v5	4	0.50000	0.40625	0.81250	0.00000
combined_hard_shift	0.10	Recover-v5	5	0.48438	0.37500	0.77419	0.00000
combined_hard_shift	0.10	Recover-v5	6	0.56250	0.42188	0.75000	0.00000
combined_hard_shift	0.10	Recover-v5	7	0.37500	0.32812	0.87500	0.00000
combined_hard_shift	0.10	Recover-v5	8	0.40625	0.34375	0.84615	0.00000
combined_hard_shift	0.10	Recover-v5	9	0.53125	0.45312	0.85294	0.00000
combined_hard_shift	0.10	Oracle	0	1.00000	0.78125	0.78125	0.00000
combined_hard_shift	0.10	Oracle	1	1.00000	0.89062	0.89062	0.01562
combined_hard_shift	0.10	Oracle	2	1.00000	0.85938	0.85938	0.00000
combined_hard_shift	0.10	Oracle	3	1.00000	0.95312	0.95312	0.00000
combined_hard_shift	0.10	Oracle	4	1.00000	0.87500	0.87500	0.03125
combined_hard_shift	0.10	Oracle	5	0.98438	0.76562	0.77778	0.03175
combined_hard_shift	0.10	Oracle	6	1.00000	0.78125	0.78125	0.01562
combined_hard_shift	0.10	Oracle	7	1.00000	0.89062	0.89062	0.01562
combined_hard_shift	0.10	Oracle	8	1.00000	0.82812	0.82812	0.01562
combined_hard_shift	0.10	Oracle	9	1.00000	0.85938	0.85938	0.00000
combined_hard_shift	0.20	Conformal filter	0	0.59375	0.45312	0.76316	0.02632
combined_hard_shift	0.20	Conformal filter	1	0.54688	0.48438	0.88571	0.02857
combined_hard_shift	0.20	Conformal filter	2	0.56250	0.50000	0.88889	0.00000
combined_hard_shift	0.20	Conformal filter	3	0.60938	0.57812	0.94872	0.02564
combined_hard_shift	0.20	Conformal filter	4	0.59375	0.54688	0.92105	0.02632
combined_hard_shift	0.20	Conformal filter	5	0.64062	0.51562	0.80488	0.00000
combined_hard_shift	0.20	Conformal filter	6	0.65625	0.62500	0.95238	0.02381
combined_hard_shift	0.20	Conformal filter	7	0.50000	0.37500	0.75000	0.00000
combined_hard_shift	0.20	Conformal filter	8	0.54688	0.53125	0.97143	0.00000
combined_hard_shift	0.20	Conformal filter	9	0.65625	0.56250	0.85714	0.02381
combined_hard_shift	0.20	Learned risk	0	0.82812	0.73438	0.88679	0.00000
combined_hard_shift	0.20	Learned risk	1	0.84375	0.68750	0.81481	0.00000
combined_hard_shift	0.20	Learned risk	2	0.79688	0.68750	0.86275	0.01961
combined_hard_shift	0.20	Learned risk	3	0.82812	0.62500	0.75472	0.01887
combined_hard_shift	0.20	Learned risk	4	0.84375	0.76562	0.90741	0.00000
combined_hard_shift	0.20	Learned risk	5	0.90625	0.78125	0.86207	0.03448
combined_hard_shift	0.20	Learned risk	6	0.85938	0.75000	0.87273	0.00000
combined_hard_shift	0.20	Learned risk	7	0.76562	0.70312	0.91837	0.00000
combined_hard_shift	0.20	Learned risk	8	0.79688	0.64062	0.80392	0.01961
combined_hard_shift	0.20	Learned risk	9	0.82812	0.70312	0.84906	0.03774
combined_hard_shift	0.20	Learned EU	0	0.78125	0.64062	0.82000	0.04000
combined_hard_shift	0.20	Learned EU	1	0.81250	0.79688	0.98077	0.00000
combined_hard_shift	0.20	Learned EU	2	0.78125	0.65625	0.84000	0.00000
combined_hard_shift	0.20	Learned EU	3	0.79688	0.60938	0.76471	0.00000
combined_hard_shift	0.20	Learned EU	4	0.78125	0.65625	0.84000	0.00000
combined_hard_shift	0.20	Learned EU	5	0.85938	0.70312	0.81818	0.00000
combined_hard_shift	0.20	Learned EU	6	0.84375	0.73438	0.87037	0.00000
combined_hard_shift	0.20	Learned EU	7	0.71875	0.54688	0.76087	0.04348
combined_hard_shift	0.20	Learned EU	8	0.79688	0.70312	0.88235	0.00000
combined_hard_shift	0.20	Learned EU	9	0.75000	0.70312	0.93750	0.00000
combined_hard_shift	0.20	CVaR recovery	0	0.71875	0.56250	0.78261	0.02174
combined_hard_shift	0.20	CVaR recovery	1	0.71875	0.54688	0.76087	0.02174
combined_hard_shift	0.20	CVaR recovery	2	0.75000	0.59375	0.79167	0.00000
combined_hard_shift	0.20	CVaR recovery	3	0.81250	0.59375	0.73077	0.03846
combined_hard_shift	0.20	CVaR recovery	4	0.75000	0.68750	0.91667	0.00000
combined_hard_shift	0.20	CVaR recovery	5	0.79688	0.65625	0.82353	0.00000
combined_hard_shift	0.20	CVaR recovery	6	0.89062	0.75000	0.84211	0.00000
combined_hard_shift	0.20	CVaR recovery	7	0.68750	0.45312	0.65909	0.02273
combined_hard_shift	0.20	CVaR recovery	8	0.71875	0.57812	0.80435	0.02174
combined_hard_shift	0.20	CVaR recovery	9	0.87500	0.71875	0.82143	0.01786
combined_hard_shift	0.20	Recover-v5	0	0.89062	0.73438	0.82456	0.00000
combined_hard_shift	0.20	Recover-v5	1	0.89062	0.76562	0.85965	0.00000
combined_hard_shift	0.20	Recover-v5	2	0.89062	0.81250	0.91228	0.01754
combined_hard_shift	0.20	Recover-v5	3	0.87500	0.71875	0.82143	0.01786
combined_hard_shift	0.20	Recover-v5	4	0.79688	0.65625	0.82353	0.00000
combined_hard_shift	0.20	Recover-v5	5	0.90625	0.78125	0.86207	0.00000
combined_hard_shift	0.20	Recover-v5	6	0.93750	0.71875	0.76667	0.00000
combined_hard_shift	0.20	Recover-v5	7	0.89062	0.73438	0.82456	0.00000
combined_hard_shift	0.20	Recover-v5	8	0.85938	0.71875	0.83636	0.00000
combined_hard_shift	0.20	Recover-v5	9	0.90625	0.71875	0.79310	0.03448
combined_hard_shift	0.20	Oracle	0	1.00000	0.78125	0.78125	0.00000
combined_hard_shift	0.20	Oracle	1	1.00000	0.89062	0.89062	0.01562
combined_hard_shift	0.20	Oracle	2	1.00000	0.85938	0.85938	0.00000
combined_hard_shift	0.20	Oracle	3	1.00000	0.95312	0.95312	0.00000

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Table 11: Seed-level fixed-risk metrics. (continued)

Split	Budget	Method	Seed	Coverage	Fixed success	Executed success	False-safe
adversarial_compound_shift	0.05	Learned risk	1	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Learned risk	2	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Learned risk	3	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Learned risk	4	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Learned risk	5	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Learned risk	6	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Learned risk	7	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Learned risk	8	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Learned risk	9	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Learned EU	0	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Learned EU	1	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Learned EU	2	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Learned EU	3	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Learned EU	4	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Learned EU	5	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Learned EU	6	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Learned EU	7	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Learned EU	8	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Learned EU	9	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	CVaR recovery	0	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	CVaR recovery	1	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	CVaR recovery	2	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	CVaR recovery	3	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	CVaR recovery	4	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	CVaR recovery	5	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	CVaR recovery	6	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	CVaR recovery	7	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	CVaR recovery	8	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	CVaR recovery	9	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Recover-v5	0	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Recover-v5	1	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Recover-v5	2	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Recover-v5	3	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Recover-v5	4	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Recover-v5	5	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Recover-v5	6	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Recover-v5	7	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Recover-v5	8	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Recover-v5	9	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.05	Oracle	0	0.73438	0.57812	0.78723	0.00000
adversarial_compound_shift	0.05	Oracle	1	0.60938	0.51562	0.84615	0.00000
adversarial_compound_shift	0.05	Oracle	2	0.54688	0.46875	0.85714	0.00000
adversarial_compound_shift	0.05	Oracle	3	0.56250	0.42188	0.75000	0.00000
adversarial_compound_shift	0.05	Oracle	4	0.73438	0.62500	0.85106	0.02128
adversarial_compound_shift	0.05	Oracle	5	0.56250	0.40625	0.72222	0.00000
adversarial_compound_shift	0.05	Oracle	6	0.62500	0.48438	0.77500	0.00000
adversarial_compound_shift	0.05	Oracle	7	0.54688	0.46875	0.85714	0.00000
adversarial_compound_shift	0.05	Oracle	8	0.51562	0.40625	0.78788	0.00000
adversarial_compound_shift	0.05	Oracle	9	0.59375	0.51562	0.86842	0.02632
adversarial_compound_shift	0.10	Conformal filter	0	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.10	Conformal filter	1	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.10	Conformal filter	2	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.10	Conformal filter	3	0.10938	0.09375	0.85714	0.00000
adversarial_compound_shift	0.10	Conformal filter	4	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.10	Conformal filter	5	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.10	Conformal filter	6	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.10	Conformal filter	7	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.10	Conformal filter	8	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.10	Conformal filter	9	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.10	Learned risk	0	0.26562	0.18750	0.70588	0.00000
adversarial_compound_shift	0.10	Learned risk	1	0.23438	0.17188	0.73333	0.00000
adversarial_compound_shift	0.10	Learned risk	2	0.29688	0.20312	0.68421	0.00000
adversarial_compound_shift	0.10	Learned risk	3	0.31250	0.25000	0.80000	0.00000
adversarial_compound_shift	0.10	Learned risk	4	0.28125	0.23438	0.83333	0.05556
adversarial_compound_shift	0.10	Learned risk	5	0.18750	0.14062	0.75000	0.00000
adversarial_compound_shift	0.10	Learned risk	6	0.40625	0.35938	0.88462	0.00000
adversarial_compound_shift	0.10	Learned risk	7	0.21875	0.20312	0.92857	0.00000
adversarial_compound_shift	0.10	Learned risk	8	0.17188	0.12500	0.72727	0.00000
adversarial_compound_shift	0.10	Learned risk	9	0.25000	0.21875	0.87500	0.00000
adversarial_compound_shift	0.10	Learned EU	0	0.25000	0.18750	0.75000	0.00000
adversarial_compound_shift	0.10	Learned EU	1	0.21875	0.20312	0.92857	0.00000
adversarial_compound_shift	0.10	Learned EU	2	0.25000	0.18750	0.75000	0.00000
adversarial_compound_shift	0.10	Learned EU	3	0.26562	0.20312	0.76471	0.05882
adversarial_compound_shift	0.10	Learned EU	4	0.26562	0.18750	0.70588	0.00000
adversarial_compound_shift	0.10	Learned EU	5	0.18750	0.17188	0.91667	0.00000
adversarial_compound_shift	0.10	Learned EU	6	0.39062	0.34375	0.88000	0.00000
adversarial_compound_shift	0.10	Learned EU	7	0.21875	0.15625	0.71429	0.00000

Table 11: Seed-level fixed-risk metrics. (continued)

Split	Budget	Method	Seed	Coverage	Fixed success	Executed success	False-safe
adversarial_compound_shift	0.10	Learned EU	8	0.12500	0.09375	0.75000	0.00000
adversarial_compound_shift	0.10	Learned EU	9	0.23438	0.21875	0.93333	0.00000
adversarial_compound_shift	0.10	CVaR recovery	0	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.10	CVaR recovery	1	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.10	CVaR recovery	2	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.10	CVaR recovery	3	0.17188	0.14062	0.81818	0.00000
adversarial_compound_shift	0.10	CVaR recovery	4	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.10	CVaR recovery	5	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.10	CVaR recovery	6	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.10	CVaR recovery	7	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.10	CVaR recovery	8	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.10	CVaR recovery	9	0.00000	0.00000	0.00000	0.00000
adversarial_compound_shift	0.10	Recover-v5	0	0.29688	0.21875	0.73684	0.00000
adversarial_compound_shift	0.10	Recover-v5	1	0.37500	0.28125	0.75000	0.00000
adversarial_compound_shift	0.10	Recover-v5	2	0.39062	0.23438	0.60000	0.00000
adversarial_compound_shift	0.10	Recover-v5	3	0.39062	0.25000	0.64000	0.04000
adversarial_compound_shift	0.10	Recover-v5	4	0.39062	0.29688	0.76000	0.00000
adversarial_compound_shift	0.10	Recover-v5	5	0.43750	0.32812	0.75000	0.03571
adversarial_compound_shift	0.10	Recover-v5	6	0.48438	0.35938	0.74194	0.03226
adversarial_compound_shift	0.10	Recover-v5	7	0.31250	0.23438	0.75000	0.00000
adversarial_compound_shift	0.10	Recover-v5	8	0.34375	0.25000	0.72727	0.00000
adversarial_compound_shift	0.10	Recover-v5	9	0.43750	0.34375	0.78571	0.00000
adversarial_compound_shift	0.10	Oracle	0	1.00000	0.81250	0.81250	0.00000
adversarial_compound_shift	0.10	Oracle	1	0.98438	0.84375	0.85714	0.01587
adversarial_compound_shift	0.10	Oracle	2	1.00000	0.85938	0.85938	0.01562
adversarial_compound_shift	0.10	Oracle	3	1.00000	0.76562	0.76562	0.00000
adversarial_compound_shift	0.10	Oracle	4	0.98438	0.81250	0.82540	0.01587
adversarial_compound_shift	0.10	Oracle	5	1.00000	0.76562	0.76562	0.01562
adversarial_compound_shift	0.10	Oracle	6	1.00000	0.78125	0.78125	0.00000
adversarial_compound_shift	0.10	Oracle	7	1.00000	0.87500	0.87500	0.01562
adversarial_compound_shift	0.10	Oracle	8	1.00000	0.78125	0.78125	0.00000
adversarial_compound_shift	0.10	Oracle	9	0.98438	0.82812	0.84127	0.01587
adversarial_compound_shift	0.20	Conformal filter	0	0.53125	0.37500	0.70588	0.02941
adversarial_compound_shift	0.20	Conformal filter	1	0.42188	0.31250	0.74074	0.00000
adversarial_compound_shift	0.20	Conformal filter	2	0.48438	0.28125	0.58065	0.00000
adversarial_compound_shift	0.20	Conformal filter	3	0.46875	0.35938	0.76667	0.00000
adversarial_compound_shift	0.20	Conformal filter	4	0.59375	0.48438	0.81579	0.00000
adversarial_compound_shift	0.20	Conformal filter	5	0.46875	0.35938	0.76667	0.00000
adversarial_compound_shift	0.20	Conformal filter	6	0.62500	0.40625	0.65000	0.05000
adversarial_compound_shift	0.20	Conformal filter	7	0.50000	0.34375	0.68750	0.00000
adversarial_compound_shift	0.20	Conformal filter	8	0.39062	0.26562	0.68000	0.04000
adversarial_compound_shift	0.20	Conformal filter	9	0.68750	0.50000	0.72727	0.04545
adversarial_compound_shift	0.20	Learned risk	0	0.81250	0.64062	0.78846	0.00000
adversarial_compound_shift	0.20	Learned risk	1	0.73438	0.54688	0.74468	0.04255
adversarial_compound_shift	0.20	Learned risk	2	0.87500	0.57812	0.66071	0.03571
adversarial_compound_shift	0.20	Learned risk	3	0.82812	0.70312	0.84906	0.01887
adversarial_compound_shift	0.20	Learned risk	4	0.81250	0.60938	0.75000	0.01923
adversarial_compound_shift	0.20	Learned risk	5	0.71875	0.57812	0.80435	0.00000
adversarial_compound_shift	0.20	Learned risk	6	0.89062	0.78125	0.87719	0.00000
adversarial_compound_shift	0.20	Learned risk	7	0.79688	0.67188	0.84314	0.01961
adversarial_compound_shift	0.20	Learned risk	8	0.65625	0.59375	0.90476	0.00000
adversarial_compound_shift	0.20	Learned risk	9	0.82812	0.65625	0.79245	0.00000
adversarial_compound_shift	0.20	Learned EU	0	0.75000	0.56250	0.75000	0.00000
adversarial_compound_shift	0.20	Learned EU	1	0.73438	0.59375	0.80851	0.00000
adversarial_compound_shift	0.20	Learned EU	2	0.81250	0.59375	0.73077	0.01923
adversarial_compound_shift	0.20	Learned EU	3	0.76562	0.59375	0.77551	0.04082
adversarial_compound_shift	0.20	Learned EU	4	0.75000	0.53125	0.70833	0.02083
adversarial_compound_shift	0.20	Learned EU	5	0.67188	0.57812	0.86047	0.02326
adversarial_compound_shift	0.20	Learned EU	6	0.79688	0.65625	0.82353	0.00000
adversarial_compound_shift	0.20	Learned EU	7	0.75000	0.60938	0.81250	0.00000
adversarial_compound_shift	0.20	Learned EU	8	0.62500	0.54688	0.87500	0.00000
adversarial_compound_shift	0.20	Learned EU	9	0.76562	0.65625	0.85714	0.00000
adversarial_compound_shift	0.20	CVaR recovery	0	0.64062	0.48438	0.75610	0.02439
adversarial_compound_shift	0.20	CVaR recovery	1	0.60938	0.37500	0.61538	0.00000
adversarial_compound_shift	0.20	CVaR recovery	2	0.62500	0.43750	0.70000	0.00000
adversarial_compound_shift	0.20	CVaR recovery	3	0.67188	0.50000	0.74419	0.02326
adversarial_compound_shift	0.20	CVaR recovery	4	0.73438	0.51562	0.70213	0.02128
adversarial_compound_shift	0.20	CVaR recovery	5	0.68750	0.54688	0.79545	0.00000
adversarial_compound_shift	0.20	CVaR recovery	6	0.78125	0.48438	0.62000	0.04000
adversarial_compound_shift	0.20	CVaR recovery	7	0.71875	0.48438	0.67391	0.02174
adversarial_compound_shift	0.20	CVaR recovery	8	0.67188	0.50000	0.74419	0.02326
adversarial_compound_shift	0.20	CVaR recovery	9	0.84375	0.65625	0.77778	0.01852
adversarial_compound_shift	0.20	Recover-v5	0	0.81250	0.60938	0.75000	0.00000
adversarial_compound_shift	0.20	Recover-v5	1	0.78125	0.59375	0.76000	0.00000
adversarial_compound_shift	0.20	Recover-v5	2	0.84375	0.60938	0.72222	0.00000
adversarial_compound_shift	0.20	Recover-v5	3	0.73438	0.50000	0.68085	0.02128
adversarial_compound_shift	0.20	Recover-v5	4	0.78125	0.57812	0.74000	0.04000

Table 11: Seed-level fixed-risk metrics. (continued)

Split	Budget	Method	Seed	Coverage	Fixed success	Executed success	False-safe
adversarial_compound_shift	0.20	Recover-v5	5	0.78125	0.60938	0.78000	0.02000
adversarial_compound_shift	0.20	Recover-v5	6	0.82812	0.67188	0.81132	0.01887
adversarial_compound_shift	0.20	Recover-v5	7	0.78125	0.64062	0.82000	0.00000
adversarial_compound_shift	0.20	Recover-v5	8	0.75000	0.54688	0.72917	0.02083
adversarial_compound_shift	0.20	Recover-v5	9	0.87500	0.64062	0.73214	0.03571
adversarial_compound_shift	0.20	Oracle	0	1.00000	0.81250	0.81250	0.00000
adversarial_compound_shift	0.20	Oracle	1	1.00000	0.85938	0.85938	0.01562
adversarial_compound_shift	0.20	Oracle	2	1.00000	0.85938	0.85938	0.01562
adversarial_compound_shift	0.20	Oracle	3	1.00000	0.76562	0.76562	0.00000
adversarial_compound_shift	0.20	Oracle	4	1.00000	0.82812	0.82812	0.01562
adversarial_compound_shift	0.20	Oracle	5	1.00000	0.76562	0.76562	0.01562
adversarial_compound_shift	0.20	Oracle	6	1.00000	0.78125	0.78125	0.00000
adversarial_compound_shift	0.20	Oracle	7	1.00000	0.87500	0.87500	0.01562
adversarial_compound_shift	0.20	Oracle	8	1.00000	0.78125	0.78125	0.00000
adversarial_compound_shift	0.20	Oracle	9	1.00000	0.84375	0.84375	0.01562

F FULL PRIOR-WORK PRESSURE TABLE

Table 12 lists the highest-ranked hostile prior-work records used to bound the novelty claim. The point is not to claim that all are direct baselines; the point is to make the surrounding field visible enough that generic uncertainty or feasibility-score claims cannot pass as novelty.

Table 12: Hostile prior-work pressure from the local deep-read pool.

Citation	Title	Year	Venue	Hostile score
(Dantam et al., 2018)	An incremental constraint-based framework for task and motion planning	2018	The International Journal of Robotics Research	35.577
(Nguyen & Vu, 2025)	XFlowMP: Task-Conditioned Motion Fields for Generative Robot Planning with Schrodinger Bridges	2025	arXiv	35.850
(Moreno et al., 2024)	Combined Task and Motion Planning via Sketch Decompositions	2024	Proceedings of the International Conference on Automated Planning and Scheduling	34.450
(Kim et al., 2022)	Representation, learning, and planning algorithms for geometric task and motion planning	2022	The International Journal of Robotics Research	33.329
(Cambon et al., 2009)	A Hybrid Approach to Intricate Motion, Manipulation and Task Planning	2009	The International Journal of Robotics Research	26.918
(Frank & Riek, 2020)	Socially intelligent task and motion planning for human-robot interaction	2020	arXiv	31.900
(Ren et al., 2021)	Extended Tree Search for Robot Task and Motion Planning	2021	arXiv	30.700
(Kim et al., 2019)	Learning to guide task and motion planning using score-space representation	2019	The International Journal of Robotics Research	30.281
(Cao & Sanderson, 1998)	AND/OR net representation for robotic task sequence planning	1998	IEEE Transactions on Systems Man and Cybernetics Part C (Applications and Reviews)	26.066
(Akbari et al., 2020)	Contingent Task and Motion Planning under Uncertainty for Human-Robot Interactions	2020	Applied Sciences	31.670
(Curtis et al., 2022)	Discovering State and Action Abstractions for Generalized Task and Motion Planning	2022	Proceedings of the AAAI Conference on Artificial Intelligence	30.592
(Yang et al., 2023)	Sequence-Based Plan Feasibility Prediction for Efficient Task and Motion Planning	2023	Robotics: Science and Systems XIX	29.217
(Li, 2019)	Predefined-Time Convergent Neural Solution to Cyclical Motion Planning of Redundant Robots Under Physical Constraints	2019	IEEE Transactions on Industrial Electronics	26.213
(Kwon & Kim, 2025)	Kinodynamic Task and Motion Planning using VLM-guided and Interleaved Sampling	2025	arXiv	31.150
(Srivastava et al., 2014)	Combined task and motion planning through an extensible planner-independent interface layer	2014	2014 IEEE International Conference on Robotics and Automation (ICRA)	29.042
(Wells et al., 2019)	Learning Feasibility for Task and Motion Planning in Table-top Environments	2019	IEEE Robotics and Automation Letters	29.030
(Vega-Brown & Roy, 2018)	Admissible Abstractions for Near-optimal Task and Motion Planning	2018	Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence	28.967
(Tosello et al., 2025)	Temporal Task and Motion Planning with Metric Time for Multiple Object Navigation	2025	Proceedings of the AAAI Conference on Artificial Intelligence	30.950
(Singh et al., 2023)	Robust feedback motion planning via contraction theory	2023	The International Journal of Robotics Research	29.917

Table 12: Hostile prior-work pressure from the local deep-read pool. (continued)

Citation	Title	Year	Venue	Hostile score
(Rajendran et al., 2022)	Unified robot task and motion planning with extended planner using ROS simulator	2022	Journal of King Saud University - Computer and Information Sciences	28.842
(Terasawa et al., 2020)	3D-CNN Based Heuristic Guided Task-Space Planner for Faster Motion Planning	2020	2020 IEEE International Conference on Robotics and Automation (ICRA)	28.809
(Lee et al., 2023)	STAMP: Differentiable Task and Motion Planning via Stein Variational Gradient Descent	2023	arXiv	29.750
(Chen et al., 2023)	AutoTAMP: Autoregressive Task and Motion Planning with LLMs as Translators and Checkers	2023	arXiv	28.750
(Li et al., 2024)	Dynamic event-triggered integrated task and motion planning for process-aware source seeking	2024	Autonomous Robots	28.750
(Hauser & Ng-Thow-Hing, 2011)	Randomized multi-modal motion planning for a humanoid robot manipulation task	2011	The International Journal of Robotics Research	25.648
(Patra et al., 2025)	Kinodynamic Motion Planning for Collaborative Object Transportation by Multiple Mobile Manipulators	2025	Journal of Mechanisms and Robotics	28.643
(Kabir et al., 2020)	Incorporating Motion Planning Feasibility Considerations during Task-Agent Assignment to Perform Complex Tasks Using Mobile Manipulators	2020	2020 IEEE International Conference on Robotics and Automation (ICRA)	28.617
(Vasilopoulos et al., 2022)	Technical Report: A Hierarchical Deliberative-Reactive System Architecture for Task and Motion Planning in Partially Known Environments	2022	arXiv	29.550

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